

xphd handbook

Exciton-phonon linewidths, symmetry and real-time dynamics from Quantum ESPRESSO, LetzElPhC and yambo — the complete workflow for planar monolayer GaN, monolayer hBN and monolayer WSe₂ (spin-orbit)

What this document is. A self-contained guide to the **xphd** package and to the calculations built on it: every command in the order it is run, what it needs, what it writes and how to tell whether its output is right; the upstream Quantum ESPRESSO, LetzElPhC and yambo steps; the conventions the code relies on; every pitfall found so far; and the state of each material. It is written so that it can be handed to a new conversation (with the package zip) and work continues without re-deriving anything.

Package state described here: 139 tests pass (3 more when yambopy is installed); 36 subcommands; 31 run scripts. Handbook compiled 30 September 2026.

For a new chat: upload this PDF and **xphd_package.zip**. Point to Section 11 (status and open items) for what remains, Section 10 (pitfalls) before changing any code, and the material sections (7–9) for the exact paths and settings. Numbers quoted as results were checked against the user's own outputs; anything not yet verified is marked *to confirm*.

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1. What xphd does, and how to install it

xphd (**X**citon-**P**honon **D**ynamics) post-processes a Quantum ESPRESSO + LetzElPhC + yambo calculation into: exciton-phonon matrix elements on the full Brillouin zone (one *archive* per exciton momentum Q); broadening-free linewidths; the real-time Boltzmann transport equation (rt-BTE) for exciton populations; the mirror (σ_h) parity of electrons, phonons and excitons and the selection rules it implies; exciton dipoles, helicity and phonon-assisted luminescence; and a set of diagnostics that decide whether the numbers can be trusted.

1.1 Install (Windows, editable)

```
# the package lives at C:\Users\risha\OneDrive\Desktop\xphd (editable install)
cd C:\Users\risha\OneDrive\Desktop\xphd
pip install -e ".[dev,plot]"
xphd --help                # lists all 36 subcommands
xphd <command> --help      # every command documents its own options
python -m pytest           # 139 pass without yambopy
```

To update: unzip a new **xphd_package.zip** over the same folder. Because the install is editable, nothing needs reinstalling. `numpy` is the only hard requirement; `matplotlib` (figures), `scipy` (BTE), and `yambopy`, `netCDF4`, `tqdm` (anything that reads yambo or LetzElPhC databases) are imported only when needed, so `--help` always works. Two kinds of tools exist:

Line continuation. Long commands are wrapped here with a trailing `\` (bash). In **PowerShell** either type the command on one line, or end each wrapped line with a backtick ``` instead. Paths in angle brackets (<SAVE>) are placeholders.

- **Subcommands** (`xphd <name> ...`): run from the command line.
- **Scripts** in `xphd/scripts/`: copied into the working folder, edited in their **SETTINGS** block and run from VS Code (Run Python File). No command line. This is the user's preferred style for every figure script.

2. The workflow at a glance

Every material goes through the same stages. Stages A–C are upstream codes; D onwards is xphd. Once the archives exist (D), the analysis branches E–I are independent.

Stage	Code	Input → output	Notes
A. Ground state	pw.x scf, then nscf → p2y	structure → <prefix>.save, yambo SAVE	nscf on the full N×N k-mesh, enough bands for the screening. Keep scf and nscf in separate directories (the nscf overwrites <prefix>.save).
B. Phonons	ph.x (ldisp, images) → q2r.x → matdyn.x	scf save → dyn files, dvscf, force constants, fine-mesh frequencies	q-grid = k-grid (N×N). loto_2d=.true. and crystal ASR in q2r and matdyn. Fine mesh 360×360 via the chunked run.sh.
C. Excitons	yambo (GW optional) → BSE at every irreducible Q	SAVE → ndb.BS_diago_Q1..Qn	Full diagonalisation (BSSmod= "d") at these sizes.
C'. El-ph	LetzElPhC	SAVE + ph.save → ndb.elph	Must use the same SAVE as the BSE (one wavefunction gauge).
D. Archives	xphd dmats, generate, check-archive, verify-archives	→ Dmats.npy, GI_ExcPh_Q0001.npz ...	Γ archive first; check-archive must print READY before the rest.
E. Linewidths	xphd matdyn, linewidth, sweep, unfold	→ hw_fine.npy, text tables, zone fields	Broadening-free triangle integration on 360×360.
F. Symmetry	xphd parity, band-parity, irreps, selection-rule; labels_from_elph.py	→ parity.npz, mode_labels_elph.npy	Parity of every state everywhere; zeros set by symmetry.
G. Optics	xphd dipoles, chirality; helicity_pl.py	→ exc_dipoles.npy, helicity-resolved PL	Complex x,y dipoles from ONE BSE.
H. Dynamics	xphd bte, bte-check; check_bte.py, valley_times.py	→ snapshots, figures	Needs every archive of the full zone.
I. Figures	plot_* scripts	→ PDFs	SETTINGS-style.

3. Conventions the code relies on

3.1 Units and indices

- **Energies** are in eV inside archives and npz files; printed tables use meV where stated. matdyn frequencies (cm^{-1}) are converted by xphd matdyn.
- **Exciton momentum index:** xphd generate iQ takes a **0-based** index over the full zone; the archive is named with iQ+1: $iQ = 0 \rightarrow \text{GI_ExcPh_00001.npz}$ (Γ).
- **States and bands** on the command line are **1-based** (--state 1 is the lowest). Band windows printed from yambo are 1-based; archive field bse_bands is a Python slice (e.g. [6, 12] = bands 7–12).
- **Fine q-list** for matdyn: 360×360 , row-major, crystal coordinates, point $i*360+j = (i/360, j/360, 0)$, with a count line first. 360 is divisible by 6, 8, 12, 18 and 24, so the fine mesh contains every source mesh.
- **Phonon branches** are energy-ordered; a branch index changes character wherever two branches cross. Selection rules are applied mode by mode, never by index.

3.2 Electron-hole pairing (critical)

yambo stores the exciton envelope with the **electron at the table index k and the hole at k+Q** (equivalently the exciton at $-Q$ in the convention electron at k, hole at $k-Q$). xphd's rep_matrix_Q shifts the **valence** rotation matrix accordingly. Only this pairing gives $|\chi| = 1$ for every state (100% against $\leq 35\%$ for the other three). The wrong pairing produced a fictitious 'mixed-parity ring' in early GaN results.

3.3 Mirror parity

- σ_h leaves every in-plane wavevector unchanged, so in any layer whose point group contains σ_h every non-degenerate state has an exact parity at every k.
- **Electrons, spinless** (GaN, hBN): diagonal of $D(\sigma_h)$ is ± 1 . **Spinors** (WSe₂ with SOC): $\sigma_h^2 = -1$ and the diagonal is $\pm i$; the label is the imaginary part, so that an exciton's parity is still label_c·label_v. At time-reversal-invariant points (Γ , M) a spinor band is a Kramers pair with opposite eigenvalues: no single label exists there.
- **Excitons:** $\chi = \sum |A|^2 p_c p_v$ generalised to the full D-matrices; must be ± 1 for every non-degenerate state.
- **Phonons:** $p = \sum_k \langle e_{\pi(k)} | M | e_k \rangle$ with $M = \text{diag}(1, 1, -1)$ and π the atom permutation of σ_h . Planar layers (GaN, hBN): $\pi = \text{identity}$ and $p = 1 - 2w_z$. WSe₂: σ_h swaps the two Se, $\pi = [0, 2, 1]$; the out-of-plane weight would mislabel A'_1 (even) and E'' (odd).
- **Selection rules:** an even phonon connects excitons of equal parity, an odd phonon excitons of opposite parity. For emission, use the axis-covariant form $\Gamma_{\text{exc}} \subset \Gamma_v \otimes \Gamma_{\text{photon}}$.

4. Data files

File	Written by	Contents	Read by
Dmats.npy	xphd dmats	single-particle rotation matrices ($n_{\text{sym}}, n_{\text{k}}, [1,] n_{\text{b}}, n_{\text{b}}$) for the BSE band window; spinor matrices include the spin rotation	generate, parity, band-parity, irreps, dipoles checks
GI_ExcPh_QXXXX.npz	xphd generate	archive for one Q: G_grid ($n_{\text{q}}, n_{\text{mod}}, n_{\text{init}}, n_{\text{final}}$) complex eV; g2_grid = $ G ^2$; Ge_grid/Gh_grid electron/hole parts; E_n_grid (states at Q); E_m_grid (states at Q+q); hw_grid; q_red, Q_red, mesh; bse_nv/bse_nc/bse_bands; elph_convention	everything downstream
hw_fine.npy	xphd matdyn	phonon energies (eV) on the 360×360 mesh, row-major	linewidth, sweep, bte, helpers
matdyn.modes	run.sh (matdyn.x)	eigenvectors on the fine mesh, one header	mode-labels, crossing_radii, plot_phonon_parity, plot_parity_bz
parity.npz	xphd parity	Q_red (full zone), chi ($n_{\text{Q}}, n_{\text{states}}$), E, iQ_ibz	selection-rule, plot_coupling_maps, crossing_radii, plots
band_parity.npz	xphd band-parity	kf, E, p, off, defined, bands (1-based band numbers of the columns)	plot_band_parity, cb_splitting, Fig_2.py
mode_labels_elph.npy	labels_from_elph.py	($n_{\text{q}}, n_{\text{mod}}$) phonon parity in the archive's order: odd → 3, even → 1, undefined → -1	selection-rule, plot_coupling_maps
exc_dipoles.npy	xphd dipoles	complex exciton dipoles (3, n) for the Q = 0 states of one BSE	helicity_pl, chirality
lw_77K.npz	xphd sweep / unfold	linewidth field over the zone	plot_linewidth_bz, bte-check
snaps_77K.npz	xphd bte --snapshots	t, F, E, Q_red, band, mesh, T (+ channel data with --channels)	check_bte, valley_times, bte_figure, bte_animate, arpes

5. Reference: all 36 subcommands

Grouped by stage. `--mesh` is derived from the archive or lattice when omitted. Most commands that read yambo take `--save` (or `--savepath`), `--bse-dir` and `--dmats`.

Command	What it does	Typical call	Writes
matdyn-split	Split the fine q-list into chunks for concurrent matdyn.x runs	<code>--n 360 --chunks 48 --prefix qlist</code>	qlist_###.txt, chunk_index.txt
matdyn-merge	Merge chunked .modes/.freq outputs with single headers, summed counts	<code>--chunks --modes-prefix --freq-prefix -o --freq --index</code>	matdyn.modes, .freq
modes-check	Verify a merged matdyn.modes: counts, duplicates, Γ acoustic residue, imaginary modes, out-of-plane weights (planar criterion)	<code>--modes --freq --n 360</code>	report
matdyn	Read the fine-mesh .freq, convert to eV, cross-check against an archive's hw_grid	<code><freq> --fine 360 --nbnd N --npz <archive> -o hw_fine.npy</code>	hw_fine.npy
dmats	Rotation matrices for the BSE band window	<code>--bse-dir --savepath --ndb-elph --nexc</code>	Dmats.npy
ibz	List the irreducible Q-points	<code>--savepath</code>	list
generate	One archive for full-zone index iQ (0-based)	<code>iQ [outdir] --nexc --savepath --ndb-elph --bse-dir --dmats --mesh n n</code>	GI_ExcPh_Q{iQ+1:04d}.npz
check-archive	Validate one archive: fields; $ G ^2=g_2$; $G_e+G_h=G$; $E_m(0)=E_n$; ASR; symmetry (time reversal and C_3 at Γ , manifold sums; OK $<1e-2$, FAIL $>5e-2$)	<code><archive> [--deg-tol 0.015] [--detail]</code>	READY / NOT READY
check	Older archive integrity checks	<code><archive></code>	report
verify-archives	All N^2 archives present and complete	<code>--dir --n</code>	report
linewidth	Golden-rule linewidth of every state at the archive's Q; broadening-free triangles on the refined mesh	<code><archive> --hw-fine --T ... --acoustic-cut 5e-4 --acoustic-model q2 --n-shell 1 [--per-mode] -o</code>	text table (meV, fs, emission and absorption)
sweep	Linewidths at every Q in a set of archives	<code>'GI_ExcPh_Q*.npz' --hw-fine --T 77 --acoustic-model q2 --n-shell 1 --workers 8 --unfold SAVE -o</code>	field npz
unfold	Symmetrise or expand an irreducible field onto the full zone	<code><field> --savepath -o</code>	field npz
interference	Split Γ into electron, hole and cross terms	<code><archive> --hw-fine --T</code>	report
channels	Channel decomposition: which final states and branches carry the width	<code><archive> --state --T --top</code>	report
frohlich	Electron/hole Fröhlich cancellation at small q	<code><archive> --nshell</code>	report
selfenergy	$\Sigma(\omega)$, quasiparticle energy, spectral function (Kramers-Kronig)	<code><archive> --state --T --hw-fine -o</code>	npz
lineshape	Full Σ matrix and absorption lineshape (off-diagonal terms)	<code><archive> --T --hw-fine --bsdb ndb.BS_diago_Q1 -o</code>	npz
localization / gauge	Real-space localisation and gauge-smoothness diagnostics	<code><archive> --hex <a in Å></code>	report
parity	$\chi(\sigma_h)$ of every state at every Q (irreducible set, unfolded by yambo's own map); with <code>--manifold-tol</code> also the σ_h matrix inside each near-degenerate manifold and the time-reversal flag of every point	<code>--save --bse-dir --dmats --nv --nstates [--manifold-tol 0.005] -o parity.npz</code>	parity.npz

Command	What it does	Typical call	Writes
band-parity	σ_h parity of the electronic bands (spinor-aware)	--save --bse-dir --dmats --nv --path G M K G -o	band_parity.npz
irreps	Irreducible representations at high-symmetry points	--save --bse-dir --dmats --nv --points K M --nstates --deg-tol	table
selection-rule	Coupling split into allowed / forbidden / undefined by parity; maps. --rotate : split by the parity eigenstates of each near-degenerate manifold (complex G), so solver-mixed states are not excluded	--archive --parity --labels --state [--by-final] [--tol] [--rotate]	PDF + numbers
mode-labels	Branch characters ZA..LO from matdyn eigenvectors (planar two-atom layers only ; prefer labels_from_elph.py)	--modes --fine --mesh --masses --archive	mode_labels.npy
decompose-parity	Why a state has a fractional χ	--parity --nv --iq --state	report
check-parity-asr	Band mirror parity and acoustic-sum-rule residue	--labels --hw-fine --nv	report
validate-planar	Checks after recomputing with a planar structure (planar only)	--masses --savepath	report
dipoles	Complex exciton dipoles from ndb.dipoles and one set of BSE eigenvectors; compared with yambo's residuals by manifold	--save --bse-dir -o [--deg-tol 0.005]	exc_dipoles.npy
chirality	Helicity and valley polarisation of a degenerate pair	--dipoles --states i j	report
coherence	k-space overlap / interference figure of merit for a pair	--pair i j	report
arpes	Exciton photoemission I(k,E), thermal or time-resolved	--temperature T --snapshots --time	npz
bte	Real-time Boltzmann transport	'GI_ExcPh_Q*.npz' --T 77 --refine 5 --hw-fine --inject optical --pump-center --pump-sigma --t-end --nt --workers --channels --snapshots	snapshots npz
bte-check	Do transport out-rates equal the linewidths?	--archives --field --hw-fine --T	report
effective-mass	Parabolic fit to a band edge from bands.x output	bands.gnu --band --center	report
mott-wannier	2D Mott-Wannier estimate of the bright-dark splitting	--Eg-dir --Eg-ind --mu-dir --mu-ind --alpha	report

--acoustic-model defaults to none in linewidth and sweep. At a band minimum the only channel is quasi-elastic acoustic absorption inside the first mesh cell, so every linewidth must pass `--acoustic-model q2 --n-shell 1`. Running with none at a minimum gives the **escape rate** out of the valley bottom instead — useful, but a different quantity.

5.1 Scripts (SETTINGS style)

Copy the script into the working folder, edit the SETTINGS block, run it.

Script	Purpose	Key settings
labels_from_elph.py	Phonon parities in the el-ph database's own mode order (permutation formula; planar and non-planar)	ELPH, ARCHIVE, SAVE (or PERM), OUT
plot_coupling_maps.py	$ G ^2$ maps, one panel per branch (2×3 or 3×3), parity contour and/or forbidden hatching; prints each panel's forbidden share	ARCHIVE, PARITY, LABELS, STATE, FINAL, OVERLAY

Script	Purpose	Key settings
crossing_radii.py	Exciton parity contour (interpolated) and the ZO crossing q_x of a branch	PARITY, MODES (360×360), N_FINE, BRANCH
helicity_pl.py	Phonon-assisted PL resolved by helicity and valley; sum-rule and time-reversal checks	ARCHIVE (Γ), DIPOLES, T_LATTICE, T_EXC, VALLEY_RADIUS
check_minimum_linewidth.py	Closed-form check of the linewidth at a band minimum (curvature, LA velocity, coupling)	ARCHIVE, HW_FINE, ALAT, MASSES, TEMPERATURES
compare_band_interp.py	Local-quadratic vs Fourier band refinement, held-out BSE points	ARCHIVE, BANDS
where_forbidden.py	What carries any forbidden coupling: phonon swaps, solver-mixed near-degenerate states, or diffuse admixture $\epsilon = (1- \chi)/2$; prints a verdict and the --tol that confirms it	ARCHIVE, PARITY, LABELS, STATE, TOL
manifold_closure.py	Are the near-degenerate manifolds whole inside the archive's states, or does the state cut split them? Verdict: more states (--nexc) or a larger --manifold-tol	PARITY (more states than the archive), NARCHIVE
window_closure.py	Is the BSE band window closed under σ_h ? Norms of $D(\sigma_h)$ band by band; a leaking band caps exciton parities below 1	DMATS, SIGMA_H_OP, FIRST_BAND, NV
qp_decompose.py	Splits GW corrections into $Z(Sx-Vxc)$ and $Z\cdot Sc$, per state and as differences between k-points: localises a k-dependent anomaly to the exchange or the screening	QP_FILE, QP_DB, KPOINTS, REFERENCE
add_osc.py	Adds the $Q = 0$ oscillator strengths (osc_Q0) that bte --inject optical needs to the Γ archive, from xphd dipoles	ARCHIVE, DIPOLES, POLARISATION
check_bte.py	Has the BTE settled, and where? Mean energy over time, and the share at the minimum's point and in each valley, against Boltzmann on the same states	BTE, T_BTE, T_CHECK, CENTERS, RADIUS
valley_times.py	Times at which each valley's population crosses 50/90/99%	SNAPS, CENTERS, RADIUS
exc_transitions.py	Band transitions and electron valley weights (Γ , K, K', Λ) of the lowest excitons at any Q	SAVE, BSE_DB, NSTATES, CENTERS
cb_splitting.py	Conduction-band spin splitting at K before/after GW; spin relation from band labels	QP_FILE, BAND_PARITY, VBM_BAND, WINDOW
soc_probe.py	Read-only diagnostic of a spin-orbit calculation (spinors, permutation, $D(\sigma_h)^2$, parities)	SAVE, BSE_DIR, DMATS, ELPH, ARCHIVE
plot_band_parity.py	Electronic bands coloured by parity	NPZ
plot_phonon_parity.py	Phonon dispersion coloured by parity (PERM for non-planar layers)	MODES, MESH_PATH, PERM
plot_parity_bz.py	Parity maps across the zone, excitons and phonons (PERM)	PARITY, MODES, BAND, PERM
plot_parity_dispersion.py	yambo interpolated exciton bands coloured by parity	INTERP, PARITY
plot_linewidth_bz.py	Linewidth over the zone	FIELD, STATE, TEMPERATURE
bte_figure.py / bte_animate.py	Population snapshots and animation	SNAPS, DISP, ALAT, PATH
plot_arpes.py	Exciton photoemission maps	NPZ
run_matdyn.sh	Chunked matdyn.x on the 360×360 list, with merged outputs	(shell variables)
run_generate.sh / .ps1	Every archive of the full zone with xphd generate, resumable; refuses unless the Γ archive is READY	N, NEXC, SAVE, ELPH, BSE, DMATS

6. Upstream settings (all materials)

xphd trusts its inputs. These settings must hold, or nothing downstream means anything.

6.1 Quantum ESPRESSO: ground state and phonons

- **Separate directories** for scf, phonons and nscf: the nscf overwrites <prefix>.save, and ph.x must read the scf ground state. Copy the scf save into each.
- **q-grid = k-grid** ($N \times N \times 1$), so every $k+q$ lies on the mesh.
- **Spin-orbit** (WSe₂): scf with `noncolin=.true.`, `lspinorb=.true.` and fully relativistic pseudopotentials; ph.x then reads SOC from the save — it needs no extra flags — and writes the spinor `dvscf` LetzElPhC requires.
- **Images** (ph.x -ni N): each image works in its own `_ph{i}` directory, so `dvscf` and patterns are spread over them (gathered by `copy.sh`, 6.3). A restart (`recover=.true.`) must keep the **same** process and pool layout.
- Errors `dirope ... error opening ./_ph0/<prefix>.q_N/<prefix>.mixd97` (many ranks at once) are file-system problems, not physics: a restart with a different layout, a node that cannot see `outdir`, or quota. Check `lfs quota`, keep `outdir` on shared scratch, and consider `reduce_io=.true.` and splitting `q` with `start_q/last_q`.

6.2 q2r.x and matdyn.x on the 360×360 mesh

```
q2r.in
&input
  fildyn = '<prefix>.dyn'
  zasr   = 'crystal'
  flfrc  = '<prefix>.fc'
  loto_2d = .true.
/

matdyn.in (namelist ONLY, ends with '/')
&input
  asr           = 'crystal'
  flfrc         = '<prefix>.fc'
  flfrq         = '<prefix>.freq'
  flvec         = 'matdyn.modes'
  loto_2d       = .true.
  q_in_cryst_coord = .true.
  q_in_band_form = .false.
  nosym         = .true.
  dos           = .false.
/
```

- `loto_2d=.true.` in **both** (the 3D non-analytic form adds a spurious LO splitting at Γ).
- `qlist.txt`: 360×360 row-major crystal coordinates with a count line (129 600 points).
- `run.sh` (Slurm): runs `matdyn.x` in 54 chunks of 2 400 points and merges `matdyn.modes`, `.freq` and `.freq.gp` with single headers, summed `nks` and a monotonic path distance; checks the namelist's closing `/`, that the force constants exist, and every count. None of `matdyn`'s outputs survives a plain `cat`.
- `matdyn.x` is serial: the parallelism is several chunks at once.

6.3 LetzElPhC

- Build `LELPH/ph.save` with `copy.sh` (run in the phonon folder): `dyn0..dynN`, `dvscf1..dvscfN`, `patterns.N.xml`, `tensors.xml`, `data-file-schema.xml` and the pseudopotentials. It searches **every** `_ph*` image (LetzElPhC's own `-pp` reads patterns only from `_ph0`), stops if one `q`'s `dvscf` differs between images (a split `q`: partial files), and flags `dvscf` files of unequal size.
- In `lelphc.in`: `ph_save_dir = ph.save` (the default is `ph_save`; the key is case-sensitive and a misspelling is silently ignored — error `Unable to open the dyn0 file`).
- **Same SAVE rule**: LetzElPhC and the BSE must read the **same** yambo SAVE; the exciton-phonon elements combine both in one wavefunction gauge. Redo both if the `nscf` is redone.

6.4 yambo: GW and BSE

- **2D:** CUTGeo= "slab z" with RIM (rim_cut). In GW, the Bruneval-Gonze terminator (GTermKind="BG", XTermKind="BG") speeds band convergence; leave VXCRLvcs at its default; SOC doubles every band count.
- **BSE:** BSEBands = the window (it fixes --nv); BSEQptR = all irreducible Q; BSSmod= "d" (full diagonalisation: the matrices are small — 2 304 for WSe₂ at 12×12, 5 184 for GaN at 24×24 — and it cannot fail to converge).
- **RIM-W in the BSE** (rim_w) broke the binding in WSe₂ (excitons 90 meV above their bare transitions). Without it they bind normally. Keep rim_cut, drop rim_w.
- **Quasiparticle corrections:** in this build (5.2.0 r22427, devel-excph), KfnQPdb= "E < ndb.QP" gave the BSE the corrections *without* the Kohn-Sham energies (report gap = the difference of the corrections). The database itself was correct. Check the report: the [K+QP] Direct Gap must equal the [X] gap plus the GW correction. Workaround: a rigid scissor % KfnQP_E.
- Maximum Residual Value in the sorted-exciton files is the normalisation of the oscillator strengths (yambo's 'residuals' are the numerators of the spectrum), **not** a convergence measure.
- At Q = 0 the long-range exchange is evaluated along one direction, splitting each E' doublet into a longitudinal and a transverse member (GaN: 2.8 meV for the first bright pair, 10 meV for the second). Each member alone is not C₃-symmetric; only the pair is.

6.5 From databases to archives (xphd)

1. Rotation matrices

```
xphd dmats --bse-dir <BSE_DIR> --savepath <SAVE> --ndb-elph <ELPH> --nexc <n>
```

What it does	Builds the single-particle rotation matrices D for every symmetry and k, restricted to the BSE band window.
Needs	SAVE, BSE databases, ndb.elph.
Writes	Dmats.npy (n_sym, n_k, [1,] n_b, n_b).
Check	Depends on the SAVE and the band window, not on the BSE solution: stays valid if only the BSE is redone.

2. The Γ archive, then the check

```
xphd generate 0 <OUT> --nexc <n> --savepath <SAVE> --ndb-elph <ELPH> --bse-dir <BSE_DIR> \
--dmats Dmats.npy --mesh N N
xphd check-archive <OUT>/GI_ExcPh_Q0001.npz [--detail]
```

What it does	Computes $G(Q=0, q)$ for every q of the mesh, every branch and every pair of states; excitons at $Q+q$ are rotated from the irreducible BSE with the D-matrices. The check validates fields, identities and symmetry.
Needs	Dmats.npy, SAVE, ndb.elph, BSE.
Writes	GI_ExcPh_Q0001.npz.
Check	READY; time reversal and C ₃ both at the 10 ⁻³ level (GaN 1.3e-3 / 3.3e-3, WSe ₂ 1.2e-3 / 1.0e-3). One operation far above the other means a rotation fault — unless --detail points to a split doublet (raise --deg-tol).

3. Every archive of the full zone (only for sweeps and transport)

```
bash run_generate.sh          # Slurm; set N, NEXC, SAVE, ELPH, BSE, DMATS
.\run_generate.ps1 -N 24 -NExc 15 -Save ... -Elph ... -Bse ... # Windows
xphd verify-archives --dir <OUT> --n <N*N>
```

What it does	Runs xphd generate for $iQ = 1 \dots N^2 - 1$, several at once (memory-bound), resumable; refuses to start unless the Γ archive is READY.
Needs	The Γ archive READY.
Writes	N^2 archives.
Check	verify-archives: all present and complete.

7. Planar monolayer GaN

Parameter	Value
Structure	planar D _{3h} honeycomb, 2 atoms (Ga, N); a = 3.23 Å in the main text (SI Table I says 3.21 — <i>reconcile</i>); cutoff 80 Ry (SI says 90 — <i>reconcile</i>)
Mesh	24×24 (576 k/q; 61 irreducible); fine mesh 360×360
BSE	bands 7–12 (bse_bands [6 12]), --nv 3, 15 states per Q
Phonons	6 branches; masses 69.723, 14.007; gan.freq; PERM = identity
Folders (D:\Data\GaN\24_planar\)	LW\LW (archives, hw_fine.npy, G.txt, K.txt, parity), BSE_EXCPH\output, BTE (snaps_77K.npz), Helicity (dipoles)
Key energies	lowest exciton at K: 3.4290 eV (dark, σ_h -odd, A ⁺); bright E' doublet at Γ : 3.8143 / 3.8171 eV; G ₀ W ₀ gap 4.78 eV indirect (K→ Γ), 4.91 eV direct
Archive of K	GI_ExcPh_Q0575.npz

GaN's archives come from an older generator (no bse_nv / elph_convention fields) but pass the symmetry check once the 10 meV split doublet (states 6–7) is grouped: time reversal 1.3×10^{-3} , C₃ 3.3×10^{-3} . No regeneration is needed.

7.1 Commands, in order

1. Fine-mesh phonons

```
xphd matdyn gan.freq --fine 360 --nbnd 6 --npz GI_ExcPh_Q0001.npz -o hw_fine.npy
```

What it does	Reads the merged 360×360 frequencies, converts to eV, checks they reproduce the archive's hw_grid where the meshes coincide.
Needs	gan.freq from run.sh.
Writes	hw_fine.npy.
Check	source-mesh agreement at the μ eV level.

2. Linewidth of the bright exciton (Γ)

```
xphd linewidth GI_ExcPh_Q0001.npz --hw-fine hw_fine.npy --T 5 77 300 --acoustic-cut 5e-4 \
--acoustic-model q2 --n-shell 1 --per-mode -o G.txt
```

What it does	Golden-rule width of every state at Γ ; energy-conserving δ integrated on 360×360 triangles; exciton energies refined by local quadratics; acoustic coupling inside the first cell by $ G ^2 2\hbar\omega = Cq^2$.
Needs	Γ archive, hw_fine.npy.
Writes	G.txt (meV, fs, emission/absorption, per branch).
Check	state 1: 36.7 meV at 5 K (99.8% emission), 51.9 meV at 77 K (12.7 fs), 150.9 meV at 300 K (43% absorption); state 2: 56.8 meV at 77 K.

3. The K minimum: width and escape

```
xphd linewidth GI_ExcPh_Q0575.npz --hw-fine hw_fine.npy --T 5 77 300 --acoustic-cut 5e-4 \
--acoustic-model q2 --n-shell 1 --per-mode -o K.txt
xphd linewidth GI_ExcPh_Q0575.npz --hw-fine hw_fine.npy --T 5 77 300 --acoustic-cut 5e-4 -o \
K_none.txt
```

What it does	With q2: the width of the minimum, from quasi-elastic LA absorption inside the valley. Without (default none): the rate of scattering beyond the first mesh cell — the escape from the valley bottom.
Needs	K archive, hw_fine.npy.
Writes	K.txt, K_none.txt.
Check	K.txt state 1: 0.087 / 10.4 (63 fs) / 50.5 meV (13 fs) at 5 / 77 / 300 K. K_none: 0.0437 meV at 77 K (≈ 15 ps). 300 K escape: read from K_none.txt.

4. Closed-form check of the K minimum

```
(script) check_minimum_linewidth.py # ARCHIVE = GI_ExcPh_Q0575.npz, ALAT = 3.23, MASSES
```

What it does	$\Gamma = (4\pi^2/\sqrt{3}) C N(\hbar\omega^*)/b^2$, $\hbar\omega^* = (\hbar c)^2/b$, from the BSE curvature, the LA velocity and the fitted coupling; compares with the code.
Needs	K archive, hw_fine.npy.
Writes	report.
Check	$b = 12.44 \text{ eV} b ^{-2}$, $\hbar\omega^* = 1.49 \text{ meV}$; closed form 10.6–11.4 meV at 77 K vs 10.4 computed. Rerun with the final ALAT for M, c_{LA} , Ξ digits.

5. Linewidths over the zone (Fig. 3 maps)

```
xphd sweep 'GI_ExcPh_Q*.npz' --hw-fine hw_fine.npy --T 77 --acoustic-cut 5e-4 --acoustic-model q2 \
--n-shell 1 --workers 8 --unfold SAVE -o lw_77K.npz
(script) plot_linewidth_bz.py # FIELD = lw_77K.npz
```

What it does	Linewidths at every irreducible Q, unfolded to the zone.
Needs	irreducible archives (or all), hw_fine.npy, SAVE.
Writes	lw_77K.npz and the map.
Check	regenerate at 300 K if Fig. 3 is shown at 300 K.

6. Exciton parity across the zone

```
xphd parity --save <SAVE> --bse-dir <BSE_DIR> --dmats Dmats.npy --nv 3 --nstates 15 -o parity.npz
```

What it does	$\chi(\sigma_h)$ of every state at every irreducible Q, unfolded with yambo's own map.
Needs	SAVE, BSE (all Q), Dmats.npy.
Writes	parity.npz.
Check	every state ± 1 ; the lowest band even inside a contour around Γ , odd outside (incl. K, M).

7. Phonon parities in the archive's mode order

```
(script) labels_from_elph.py # ELPH, ARCHIVE = GI_ExcPh_Q0001.npz, SAVE
```

What it does	Parity of every (q, mode) from the el-ph database's own eigenvectors (matdyn orders nearly degenerate modes differently at ~5% of points).
Needs	ndb.elph, archive, SAVE.
Writes	mode_labels_elph.npy.
Check	'planar layer'; frequencies match the archive (same order).

8. The selection rule, quantified

```
xphd selection-rule --archive GI_ExcPh_Q0001.npz --parity parity.npz --labels mode_labels_elph.npy \
--state 1 --by-final
```

What it does	Splits the coupling from the initial state into allowed, forbidden and undefined by parity.
Needs	archive, parity.npz, labels.
Writes	numbers and maps.
Check	forbidden/allowed = 5×10^{-7} (zero to numerical precision).

9. Labels at K and M; band parity

```
xphd irreps --save <SAVE> --bse-dir <BSE_DIR> --dmats Dmats.npy --nv 3 --points K M --nstates 4
xphd band-parity --save <SAVE> --bse-dir <BSE_DIR> --dmats Dmats.npy --nv 3 --path G M K G -o \
bandparity.npz
```

What it does	Irreducible representations of the lowest states; parity of the electronic bands.
Needs	SAVE, BSE, Dmats.npy.
Writes	tables; band_parity.npz.
Check	lowest K exciton A" (odd). Band edges: VBM at K is N p _z (odd), CBM at Γ is Ga s (even).

10. Parity contour and the ZO crossing

```
(script) crossing_radii.py # PARITY = parity.npz, MODES = 360x360 matdyn.modes, BRANCH = 4
```

What it does	Interpolated crossing of the lowest even and odd exciton bands; where branch 4 turns from ZO (odd) to in-plane (even).
Needs	parity.npz, matdyn.modes.
Writes	radii along Γ -K and Γ -M.
Check	contour 0.256 b (Γ -K), 0.265 b (Γ -M). q _x : <i>pending</i> (second half of the run).

11. Figures: coupling maps and parity

```
(script) plot_coupling_maps.py # STATE = 1, OVERLAY = contour (and 'both' as a check)
(script) plot_band_parity.py, plot_phonon_parity.py, plot_parity_bz.py, plot_parity_dispersion.py
```

What it does	Six-panel coupling maps with the parity contour; parity of bands, phonons and excitons.
Needs	archive, parity.npz, labels, band_parity.npz, matdyn.modes.
Writes	PDFs.
Check	each panel's forbidden share ~0 with OVERLAY = both.

12. Exciton dipoles and the helicity-resolved sideband

```
xphd dipoles --save ../LW/LW/SAVE/ --bse-dir ../BSE_EXCPH/output -o exc_dipoles.npy --deg-tol 0.005
(script) helicity_pl.py # ARCHIVE = GI_ExcPh_Q0001.npz, Dipoles = exc_dipoles.npy
```

What it does	Complex x, y, z dipoles of the Q = 0 states from one BSE; then phonon-assisted PL resolved by circular polarisation and by valley.
Needs	SAVE, BSE (ndb.dipoles), Γ archive.
Writes	exc_dipoles.npy, helicity_pl.pdf/.npz.
Check	yambo residuals agree per manifold (ratio $\approx 3.47 \times 10^7$ for split pairs summed); satellite 3.4128 eV (16.2 meV below the dark exciton); P = -1.0000 (K), +1.0000 (K'); sum rule 3.7e-16; K/K' 0.9988.

13. Real-time transport (Fig. 4)

```
xphd bte 'GI_ExcPh_Q*.npz' --T 77 --refine 5 --hw-fine hw_fine.npy --inject optical \
--pump-center <E_bright> --pump-sigma 0.018 --t-end 2000 --nt 200 --workers 14 \
--channels --snapshots snaps_77K.npz
(scripts) check_bte.py, valley_times.py, bte_figure.py, bte_animate.py
xphd bte-check --archives 'GI_ExcPh_Q*.npz' --field lw_77K.npz --hw-fine hw_fine.npy --T 77
```

What it does	Propagates the populations of all states at all Q with the full rate matrix; detailed balance symmetrised; rates refined by row rescaling.
Needs	all 576 archives, hw_fine.npy.
Writes	snapshots; figures.
Check	end state 4.26 meV vs Boltzmann 4.09 meV at 77 K; valleys: 35% at 30 fs, 50% at 41, 90% at 98, 99% at 172 fs. <i>Confirm the pump settings used</i> (main text: resonant; SI: $\sigma = 18$ meV).

14. Diagnostics, as needed

```
xphd channels GI_ExcPh_Q0001.npz --state 1 --T 77
xphd interference GI_ExcPh_Q0001.npz --hw-fine hw_fine.npy --T 77
xphd frohlich GI_ExcPh_Q0001.npz
xphd lineshape GI_ExcPh_Q0001.npz --T 77 --hw-fine hw_fine.npy --bsdb <BSE_DIR>/ndb.BS_diago_Q1 \
-o eps2.npz
```

What it does	Where the width comes from; electron/hole interference; Fröhlich cancellation; off-diagonal self-energy.
Needs	archives.
Writes	reports.
Check	polar branches: nothing from the Γ bright state by emission (parity); check absorption at 300 K from the per-mode columns.

8. Monolayer hBN (the bright-minimum contrast case)

Parameter	Value
Structure	planar D _{3h} , 2 atoms (B, N); a from the relaxed scf.out (≈ 2.50 Å)
Mesh	24×24 (576), fine 360×360
BSE	bse_bands [1 7], --nv 3, 15 states
Phonons	6 branches; masses 10.81, 14.007; bn.freq; PERM = identity
Folders (D:\Data\hBN\, from the earlier checklist — confirm)	../phonons/SAVE, ../phonons/ndb.elph, ../BSE/output_all, archives/
Established	lowest exciton at Γ , 5.475 eV, bright (E' doublet split 0.2 meV); K at 5.62 eV, 150 meV above (the opposite ordering to GaN); PL: LO/TO sidebands of the bright exciton; el-ph elements single precision ($ G_e+G_h-G / G = 1.7 \times 10^{-7}$); 18×18 $\rightarrow$ 24×24 lowers the exciton by 200–280 meV (not converged in absolute terms)

Earlier hBN parity numbers (even at K, +0.96; odd pocket at M, −0.96) and the forbidden fraction 0.017 predate two fixes — the electron-hole pairing and the phonon labels from the el-ph database.

Recompute them with the commands below before quoting anything.

8.1 Commands, in order

1. Check the BSE energies went in correctly (this yambo build)

```
Select-String -Path <BSE folder>\r-* -Pattern "[K\+QP\] Direct Gap|[X\] Direct Gap"
```

What it does	Confirms the quasiparticle correction was added to the Kohn-Sham energies (see 6.4).
Needs	the BSE report.
Writes	—
Check	[K+QP] gap = [X] gap + GW correction.

2. Archives (if not current)

```
xphd dmats --bse-dir ../BSE/output_all --savepath ../phonons/SAVE --ndb-elph ../phonons/ndb.elph \
--nexc 15
xphd generate 0 archives --nexc 15 --savepath ../phonons/SAVE --ndb-elph ../phonons/ndb.elph \
--bse-dir ../BSE/output_all --dmats Dmats.npy --mesh 24 24
xphd check-archive archives/GI_ExcPh_Q0001.npz
```

What it does	As in 6.5.
Needs	SAVE, ndb.elph, BSE.
Writes	Dmats.npy, archives.
Check	READY; symmetry at the 10^{-3} level.

3. Parity, labels, selection rule

```
xphd parity --save ../phonons/SAVE --bse-dir ../BSE/output_all --dmats Dmats.npy --nv 3 --nstates \
15 -o parity.npz
(script) labels_from_elph.py # ELPH = ../phonons/ndb.elph, ARCHIVE, SAVE = ../phonons/SAVE
xphd selection-rule --archive archives/GI_ExcPh_Q0001.npz --parity parity.npz \
--labels mode_labels_elph.npy --state 1 --by-final
```

What it does	As for GaN.
Needs	as for GaN.
Writes	parity.npz, labels, numbers.
Check	forbidden/allowed near zero; the lowest exciton even at Γ .

4. Labels and band parity

```
xphd irreps --save ../phonons/SAVE --bse-dir ../BSE/output_all --dmats Dmats.npy --nv 3 --points \
K M --nstates 4
xphd band-parity --save ../phonons/SAVE --bse-dir ../BSE/output_all --dmats Dmats.npy --nv 3 -o \
band_parity.npz
```

What it does	Irreps at K and M; parity of the band edges (both π -like at K: class I).
Needs	as above.
Writes	tables, band_parity.npz.
Check	band edges both odd (p_z) at K; lowest exciton even and bright.

5. Negative control for the helicity

```
xphd dipoles --save ../phonons/SAVE --bse-dir ../BSE/output_all -o exc_dipoles.npy --deg-tol 0.005
(script) helicity_pl.py # hBN  $\Gamma$  archive
```

What it does	A bright minimum at Γ has no valley-locked sideband.
Needs	SAVE, BSE, Γ archive.
Writes	helicity_pl.pdf.
Check	$ P $ near zero / no valley satellite.

6. Linewidths (optional)

```
xphd matdyn bn.freq --fine 360 --nbnd 6 --npz archives/GI_ExcPh_Q0001.npz -o hw_fine.npy
xphd linewidth archives/GI_ExcPh_Q0001.npz --hw-fine hw_fine.npy --T 5 77 300 --acoustic-cut 5e-4 \
--acoustic-model q2 --n-shell 1 --per-mode -o Gamma.txt
(script) check_minimum_linewidth.py # ARCHIVE =  $\Gamma$  archive, ALAT from scf, MASSES = [10.81, 14.007]
```

What it does	The bright Γ state is hBN's minimum: its width is set by in-valley acoustic absorption (q2 essential).
Needs	Γ archive, hw_fine.npy.
Writes	Gamma.txt.
Check	closed form agrees with the code.

9. Monolayer WSe₂ with spin-orbit coupling

Parameter	Value
Structure	1H WSe ₂ , D _{3h} , 3 atoms (W, Se, Se); a = 6.27846 bohr (3.322 Å); σ_h maps atoms [0,1,2] → [0,2,1] (operation 6 of 24)
Spin	spinors (2 components), 62 electrons → 62 occupied bands; time reversal on
Mesh	12×12 (144; 19 irreducible); fine 360×360 (factor 30)
Upstream	prefix wse; nscf 200 bands; phonons with images (19); GW bands 1-200, ndb.QP for 59-66 at all 19 k
BSE	bands 59-66 (--nv 4, nc 4); BSSmod= "d"; scissor % KfnQP_E 1.357578 1 1 , KfnQPdb= "none"; no rim_w (keep rim_cut)
Archive	10 states per Q (--nexc 10); 9 branches
Folders (D:\Data\WSe2\)	LELPH\SAVE (SAVE for BSE and LetzElPhC), LELPH\ph.save, GW\output, BSE\output (current BSE), SLEPC (test runs), EXCPH (Dmats.npy, archives, parity.npz), phonon folder with _ph0.._ph18, copy.sh; matdyn inputs in wse2_matdyn

9.1 What the upstream checks established

- **GW is right:** every band 59-66 corrected at every k; at K the conduction splitting E(64)–E(63) is +37.2 meV (DFT) and +22.2 meV (GW); band 63 has the opposite spin to the valence maximum (WSe₂ ordering). Corrections at K: band 62 –0.4709 eV, 63 +0.8867 eV, 64 +0.8716 eV; the scissor 1.357578 eV is the gap correction at K.
- **The database route failed in the BSE:** with KfnQPdb the BSE gap at K was 1.3576 eV (the corrections alone), which reversed bands 63 and 64 and put the bright exciton below the dark one. Every exciton was 1.2573 eV below the scissor run's — the Kohn-Sham gap.
- **RIM-W broke the binding;** without it: dark pair 1.3375 / 1.3407 eV, bright A doublet 1.3907 eV (dark 53 meV below bright: 37 meV Kohn-Sham splitting + ~16 meV exchange); bright B 1.8038 eV (A-B 0.41 eV); binding ~1.27 eV (overbound by the coarse mesh; absolute energies not converged).
- **SOC probe:** $D(\sigma_h)^2 = -1.0000$ (spin rotation included); every Q = 0 exciton ± 1 ; spin-forbidden states odd, spin-allowed even. Phonon parity by permutation ± 1 to 7×10^{-16} (the out-of-plane formula would mislabel 34.4%). Each band is exactly 50% even: spin-valley locking.

9.2 The full build, in order

From D:\Data\WSe2\EXCPH, with ../LELPH/SAVE, ../LELPH/ndb.elph and ../BSE/output (the BSE without RIM-W, with the scissor). Copy the scripts you use into the folder.

1. Phonons on the 360×360 mesh (cluster, phonon folder)

```
q2r.x < q2r.in > q2r.out          # wse.dyn0..dyn19 -> wse.fc
sbatch run.sh                     # 54 chunks -> wse.freq, matdyn.modes
xphd modes-check --modes matdyn.modes --freq wse.freq --n 360
```

What it does	Force constants, then frequencies and eigenvectors on 129 600 q-points (inputs in wse2_matdyn).
Needs	the 19 dyn files.
Writes	wse.freq, matdyn.modes (copy both to EXCPH).
Check	129 600 points, no duplicates, no imaginary modes; the out-of-plane line warns — expected for a non-planar layer.

2. Rotation matrices and the Γ archive

```
mkdir archives
xphd dmats --bse-dir ../BSE/output --savepath ../LELPH/SAVE --ndb-elph ../LELPH/ndb.elph --nexc 10
xphd generate 0 archives --nexc 10 --savepath ../LELPH/SAVE --ndb-elph ../LELPH/ndb.elph \
  --bse-dir ../BSE/output --dmats Dmats.npy --mesh 12 12
xphd check-archive archives/GI_ExcPh_Q0001.npz
```

What it does	As in 6.5.
Needs	SAVE, ndb.elph, BSE.
Writes	Dmats.npy, archives/GI_ExcPh_Q0001.npz.
Check	READY; $E_n = 1.3375, 1.3407, 1.3907, 1.3907 \dots$; time reversal and $C_3 \approx 1 \times 10^{-3}$.

3. Every archive (144)

```
pwsh .\run_generate.ps1 -N 12 -NExc 10 -Throttle 4 -Out archives -Save ../LELPH/SAVE -Elph \
  ../LELPH/ndb.elph -Bse ../BSE/output
xphd verify-archives --dir archives --n 144
```

What it does	iQ = 1 ... 143, resumable (PowerShell 7 for -Throttle above 1).
Needs	the READY Γ archive.
Writes	144 archives.
Check	all present and complete.

4. Fine-mesh phonons in eV

```
xphd matdyn wse.freq --fine 360 --nbnd 9 --mesh 12 12 --npz archives/GI_ExcPh_Q0001.npz -o \
  hw_fine.npy
```

What it does	As for GaN; refinement factor 30.
Needs	wse.freq, the Γ archive.
Writes	hw_fine.npy.
Check	agreement with hw_grid on the source mesh.

5. Symmetry

```
xphd parity --save ../LELPH/SAVE --bse-dir ../BSE/output --dmats Dmats.npy --nv 4 --nstates 10 \
  --manifold-tol 0.005 -o parity.npz
(script) labels_from_elph.py # ELPH, ARCHIVE = archives/GI_ExcPh_Q0001.npz, SAVE = ../LELPH/SAVE
xphd selection-rule --archive archives/GI_ExcPh_Q0001.npz --parity parity.npz --labels \
  mode_labels_elph.npy --state 1 --by-final --out sel_s1.pdf
xphd selection-rule ... --state 1 --rotate --out sel_s1_rot.pdf
xphd selection-rule ... --state 1 --tol 0.96 --out sel_s1_t096.pdf (and the same three for \
  --state 3)
(script) where_forbidden.py # STATE = 1, then 3
xphd band-parity --save ../LELPH/SAVE --bse-dir ../BSE/output --dmats Dmats.npy --nv 4 -o \
  band_parity.npz
```

What it does	Parity of excitons, phonons and bands; the selection rule from the odd dark (1) and even bright (3) states.
Needs	archives, SAVE, BSE, Dmats, ndb.elph.
Writes	parity.npz, labels, selection-rule PDFs, band_parity.npz.
Check	Established: forbidden/allowed 1.4×10^{-4} (state 1) and 1.0×10^{-3} (state 3) at --tol 0.5, entirely from one solver-mixed pair (states 7, 8, 0.13 meV apart, $ x = 0.915, 0.951$, one star of q); at --tol 0.96: 1.8×10^{-7} and 1.4×10^{-7} (GaN 5×10^{-7}), with 22% / 49% of the coupling then undefined. --rotate splits by the parity eigenstates of each near-degenerate manifold instead: nothing excluded. Run for state 1: 6.3×10^{-6} (22× below per-state), undefined 5.9%, but only 70% of eigen-parities within 0.02 of ± 1 — some manifolds incomplete. manifold_closure.py (parity with --nstates 16) decides whether the 10-state cut causes it; if so, generate the archives with --nexc 16.

6. Linewidths (the dark Γ exciton is the zone minimum)

```
xphd linewidth archives/GI_ExcPh_Q0001.npz --hw-fine hw_fine.npy --T 5 77 300 --acoustic-cut 5e-4 \
--acoustic-model q2 --n-shell 1 --per-mode -o G_WSe2.txt
xphd linewidth archives/GI_ExcPh_Q0001.npz --hw-fine hw_fine.npy --T 5 77 300 --acoustic-cut 5e-4 \
-o G_WSe2_none.txt
(script) check_minimum_linewidth.py # ARCHIVE  $\Gamma$ , STATE = 0, FINAL = 0, ALAT = 3.3224, MASSES = \
[183.84, 78.96, 78.96], REFINE = 30
xphd linewidth <K archive> ... -o K_WSe2.txt # the K-K' exciton; find its file by Q_red = \
(1/3, 1/3)
xphd sweep "archives/GI_ExcPh_Q*.npz" --hw-fine hw_fine.npy --T 77 --acoustic-cut 5e-4 \
--acoustic-model q2 --n-shell 1 --workers 8 --unfold ../LELPH/SAVE -o lw_77K.npz
```

What it does	Widths of every state at Γ (dark minimum: q2 essential) and at K; escape rate (default none); the zone map.
Needs	archives, hw_fine.npy.
Writes	text tables; lw_77K.npz.
Check	closed form agrees with the code. The Γ valley is smaller than one mesh cell (0.083 b): the width rests on the q2 model.

7. Dipoles, optical injection, transport

```
xphd dipoles --save ../LELPH/SAVE --bse-dir ../BSE/output -o exc_dipoles.npy --deg-tol 0.005
(script) add_osc.py # ARCHIVE = archives/GI_ExcPh_Q0001.npz, Dipoles = exc_dipoles.npy
xphd bte "archives/GI_ExcPh_Q*.npz" --T 77 --refine 5 --hw-fine hw_fine.npy --inject optical \
--pump-center 1.3907 --pump-sigma 0.010 --t-end 2000 --nt 200 --workers 8 --channels \
--snapshots snaps_77K.npz
(script) check_bte.py, valley_times.py # CENTERS = {"Gamma": [0, 0], "K": [1/3, 1/3], "K'": \
[2/3, 2/3]}
xphd bte-check --archives "archives/GI_ExcPh_Q*.npz" --field lw_77K.npz --hw-fine hw_fine.npy --T 77
```

What it does	Complex dipoles (checked against yambo's residuals); in-plane oscillator strengths into the Γ archive (generate does not store them); transport from the bright A doublet.
Needs	BSE (ndb.dipoles), all archives, hw_fine.npy.
Writes	exc_dipoles.npy, snaps_77K.npz.
Check	residual check passes; dark states get no in-plane strength; end state near Boltzmann at 77 K; populations at Γ (dark) and K (K-K').

8. Figures

```
(script) plot_coupling_maps.py # ARCHIVE  $\Gamma$ , STATE = 1 (then 3), OVERLAY = contour; once 'both'
(script) plot_band_parity.py; plot_phonon_parity.py and plot_parity_bz.py with PERM = [0, 2, 1]; \
plot_linewidth_bz.py
(script) helicity_pl.py # optional: sidebands of emitters in the K and K' discs (the K-K' exciton)
```

What it does	3×3 coupling maps with the parity contour; parity of bands, phonons, excitons; linewidth map.
Needs	the files above.
Writes	PDFs.
Check	helicity_pl.py excludes $q = 0$, so the Γ -phonon replicas of the dark Γ exciton are not in it.

10. Pitfalls found, and what fixed them

Symptom	Cause	Fix / status
A ring of fractional parity (0.26–0.38 [b]) in GaN	wrong electron-hole pairing in the exciton rotation	valence matrix shifted (rep_matrix_Q); every state ± 1
~1.6% 'forbidden' coupling	phonon labels from matdyn, ordered differently from the el-ph modes at ~5% of points	labels_from_elph.py $\rightarrow 5 \times 10^{-7}$
K linewidth 34% off, spurious anisotropy	global Fourier interpolation of energy-ordered exciton bands rings at crossings	local quadratic refinement (default --band-interp local)
K minimum width ~0.04 meV instead of 10 meV	acoustic channel inside the first cell missing	--acoustic-model q2 --n-shell 1 (default is none)
Old compute_parity.py: 'could not be unfolded'	hard-coded irreducible Q table from GaN	xphd parity (yambo's own map)
yambopy 'open-shell' error / shifted conduction block reading ndb.dipoles	yambopy miscounts DipBandsAll bands when the first band is not 1	xphd dipoles reads the full file and selects the window
Helicity-resolved PL identical for $\sigma^{\pm}$	valleys summed (time reversal) and/or dipoles from two BSE runs	one BSE's complex x,y dipoles; valley-resolved sums
WSe ₂ bright exciton below the dark one	this yambo build applied ndb.QP without the KS energies	rigid scissor; check [K+QP] gap in the report
WSe ₂ excitons above their bare transitions	RIM-W in the BSE kernel	drop rim_w (keep rim_cut)
'Maximum Residual Value = 0.2' read as non-convergence	it normalises oscillator strengths	no action; full diagonalisation anyway
ph.x diroprn errors on many ranks	restart with another layout / node visibility / quota	same layout on recover; shared outdir; start_q/last_q
LetzElPhC: 'Unable to open the dyn0 file'	ph_save_dir not set (default ph_save)	ph_save_dir = ph.save (lower case)
Non-planar phonons mislabelled	out-of-plane weight assumes every atom on the mirror	permutation formula; PERM in scripts; mode-labels refuses 3 atoms
WSe ₂ forbidden residue 10^{-4} – 10^{-3} in states with $ \chi \approx 0.9$	a 0.13 meV opposite-parity pair returned mixed by the solver	--tol 0.96 confirms (1.8×10^{-7}); --rotate (manifold eigenstates) excludes nothing
GaN C ₃ check FAIL (0.20) with time reversal OK	doublet split 10 meV by long-range exchange; members individually anisotropic to small-q LO	--deg-tol default 15 meV; --detail names split partners
Spinor band parity 'undetermined' at Γ , M	Kramers pairs have opposite σ_h eigenvalues	physical; exciton parity unaffected
matdyn outputs corrupted by cat	headers and cumulative distances per chunk	run.sh / matdyn-merge
Emission rule not axis-invariant	$\Gamma_v \subset \Gamma_{\text{exc}}^* \otimes \Gamma_{\text{photon}}$	use $\Gamma_{\text{exc}} \subset \Gamma_v \otimes \Gamma_{\text{photon}}$

11. Status and open items (30 September 2026)

11.1 GaN paper (PRL target)

Established and checked: exact σ_h selection rule (5×10^{-7}); the lowest exciton at K is σ_h -odd (A'') on a band disconnected from the bright one; single-branch (E''_1 , 16.2 meV), fully circularly polarised, valley-locked sideband ($P = \mp 1.0000$ at K/K'; absent partner E''_2); linewidths (Γ : 36.7 / 51.9 / 150.9 meV at 5 / 77 / 300 K; K: 0.087 / 10.4 / 50.5 meV) with a closed-form check; escape 0.044 meV (15 ps) at 77 K; BTE funnelling 50/90/99% by 41/98/172 fs, detailed balance 4.26 vs 4.09 meV; parity contour 0.256/0.265 |b|; archive symmetry verified.

Open:

- q_x for the fourth branch (crossing_radii.py second half); check it lies outside the contour.
- 300 K per-branch breakdown of the Γ width (G.txt per-mode) and whether polar branches contribute by absorption at 300 K; 300 K escape rate (K_none.txt); the temperature of the 707 meV state-4 value.
- Fig. 3 regenerated at 300 K; Figs. 6, 7, 11 of the SI regenerated with the current runs; Fig. 1 coloured by parity with σ/π labels.
- SI placeholders (compare_band_interp rms, $M/c_{LA}/\Xi$ digits with the final ALAT, off-diagonal lineshape numbers, BTE pump settings); 80 vs 90 Ry and 3.23 vs 3.21 Å.
- Bibliography fixes; Supplemental Material wording ('[URL will be inserted by publisher]').

11.2 hBN

- 300 K BTE: the end state is detailed-balanced (emission = absorption for every branch) but spread over the Γ valley's neighbours, as Boltzmann at 300 K is on a 24×24 mesh. Check with check_bte.py (CENTERS = Γ). For the contrast with GaN, run hBN at GaN's 77 K.
- Recompute parity, labels and the selection rule with the fixed code (Section 8).
- Irreps at K; helicity negative control; the classification-table row.
- Check the BSE report's [K+QP] gap (the quasiparticle-application fault in this build).

11.3 WSe₂

- **Exciton ordering — the GW is faulty.** The reference (Chen, Sangalli, Bernardi) has an indirect K→Q quasiparticle gap and the lowest exciton at M (K-Q). Our corrections move the Q conduction band 0.41 eV more than K's (DFT: Q 64 meV above K; GW: 473 meV above), the Γ valence band 0.50 eV more than K's, and stretch the band offsets about twofold — not credible. The BSE's rigid scissor (K's correction) kept DFT's +64 meV, which still leaves M (1.4003 eV) above the dark Γ state (1.3375). If Q sat about 0.1 eV below K, the K-Q exciton would fall of order 0.16 eV, below Γ . *Next:* qp_decompose.py (exchange vs correlation); suspects RIM-W in the GW and only 200 spinor bands (Xb, Scb 1–200); then a corrected GW, and a BSE that applies k-dependent corrections (KfnQPdb works only in a build without this branch's fault). Hold the 144 archives until then.
- Selection rule settled at --tol 0.96 (1.8×10^{-7} / 1.4×10^{-7}), the default-tolerance residue being one solver-mixed pair. The manifold rotation is now implemented (parity --manifold-tol, selection-rule --rotate): rerun parity with it and report the rotated numbers, which exclude nothing.
- Full build (9.2): 144 archives, hw_fine, linewidths, sweep, dipoles + add_osc, BTE, figures.
- Decide SI inclusion: symmetry demonstration with caveats (rigid scissor, no RIM-W, 12×12 , absolute energies unconverged).
- Report the two yambo faults (QP application; RIM-W in the BSE) to the developers.

11.4 Package

- 139 tests (+3 with yambopy). Manifold rotation: parity --manifold-tol stores in-manifold σ_h matrices and time-reversal flags; selection-rule --rotate splits the complex coupling by their eigen-parities (conventions in xphd/manifold.py; tested on mixed pairs with time-reversed partners). where_forbidden.py prints a verdict with the confirming --tol; add_osc.py supplies the oscillator strengths optical injection needs (generate does not store them); check_bte.py and valley_times.py take CENTERS (add Γ for WSe₂). Symmetry thresholds in check-archive calibrated on GaN and WSe₂ (OK < 10^{-2} , FAIL > 5×10^{-2}).
- run_generate.sh/.ps1 now call xphd generate with explicit paths and refuse without a READY Γ archive.
- plot_phonon_parity.py and plot_parity_bz.py take PERM for non-planar layers.

Part II — Methods and derivations

The physics and numerics behind the xphd results: every formula the code evaluates, how it is derived, and how each was checked

Part I (pages 1–24) explains the package: the commands, scripts, data files and material walkthroughs. Part II collects the methods behind them. Each section states the quantity, derives the working formula, describes how it is evaluated on the mesh, and gives the checks that established it, with the numbers obtained for GaN, hBN and WSe₂. Results that are still pending are marked [in red]. The references at the end carry their verification status in the accompanying partII.bib: entries marked VERIFY there should be checked before they are cited in a paper.

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Errata to Part I (4 October 2026)

1. **No KfnQPdb fault.** Sections 6.4, 9.1, 9.2 and the pitfalls table of Sec. 10 state that Yambo's KfnQPdb applied the quasiparticle corrections without the Kohn-Sham energies. That is wrong. With `ExtendOut, o-*.qp` lists K-point, Band, E_o, E, E-E_o, V_{xc}, V_{nlxc}, Sc|E_o, ...: the fourth column is the quasiparticle energy E , and reading it as $E - E_o$ produced the false diagnosis. KfnQPdb applies `ndb.QP` correctly; the report's `[K+QP]` gap is the true quasiparticle gap.
2. **The numbers of Sec. 9.1.** The values quoted there as corrections at **K** ($-0.4709, +0.8867, +0.8716$ eV for bands 62, 63, 64) are the quasiparticle energies. The corrections of that (RIM-W) GW are $-0.471, -0.373$ and -0.425 eV: it opened the gap at **K** by only $+0.098$ eV (the report's 1.3576 eV) and put band 64 below band 63 by 15 meV (not $+37.2$ to $+22.2$ meV). That ordering, not a code fault, is why the BSE with the database put the bright exciton lowest.
3. **The scissor.** The 1.357578 eV scissor of the WSe₂ BSE runs is that GW's quasiparticle gap at **K**, not its correction; those runs used a gap of 2.617 eV at **K**. The GW without RIM-W gives a correction of $+1.181$ eV (gap 2.440 eV for $62 \rightarrow 63$, 2.422 eV for $62 \rightarrow 64$).
4. **RIM-W, scope.** Sections 6.4 and 10 of Part I advise dropping `rim_w` in general. That holds for WSe₂ only: there it broke the BSE kernel and the GW (gap correction at **K** $+0.10$ eV with it, $+1.18$ eV without), whereas GaN and hBN ran correctly with RIM-W. The cause is not isolated (spinors, the 12×12 mesh and the RIM-W settings all differed). Check the plausibility of the gap correction and of the binding energy; if in doubt, compare a run without `rim_w`.
5. **Tools.** `cb_splitting.py` read `o-*.qp` by position and misread `ExtendOut` files; it now reads columns by header name, as `qp_decompose.py` does, and reports band edges and the direct or indirect gap (Sec. 19).

12. Notation and conventions

Lattice. The hexagonal reciprocal basis $\mathbf{b}_1, \mathbf{b}_2$ encloses 60° ; reduced coordinates (q_1, q_2) give $\mathbf{q} = q_1 \mathbf{b}_1 + q_2 \mathbf{b}_2$, and lengths are quoted in units of $|\mathbf{b}| = |\mathbf{b}_1|$. In these units the Brillouin zone has area $A_{\text{BZ}} = \sqrt{3}/2$, $\mathbf{K} = (1/3, 1/3)$, $\mathbf{K}' = (2/3, 2/3)$ and the **M** points are the edge midpoints, e.g. $(1/2, 0)$. The source mesh is $N \times N$ ($N = 24$ for GaN and hBN, 12 for WSe₂), with spacing $h = 1/N$; the refined mesh used for integration is 360×360 .

Electron-phonon elements. Computed by LetzElPhC [1] in the Yambo convention, $g_{n'\nu}(\mathbf{k}, \mathbf{q}) = \langle \psi_{n'\mathbf{k}} | \Delta_{\nu\mathbf{q}} V | \psi_{n,\mathbf{k}-\mathbf{q}} \rangle$, and converted to the standard convention before projection. The phonon displacement normalisation is that of DFPT [2, 3].

Exciton envelopes. The BSE is solved with Yambo [4] at every irreducible **Q**: for GaN with the subspace-iteration solver of SLEPc [5], for WSe₂ by full diagonalisation (matrices of dimension 2304). We write $A_{v\mathbf{c}\mathbf{k}}^{SQ}$ with the electron at $\mathbf{k}$ and the hole at $\mathbf{k} - \mathbf{Q}$. Yambo's BSE tables store the electron at the table index $\mathbf{k}$ and the hole at $\mathbf{k} + \mathbf{Q}$, i.e. the exciton at $-\mathbf{Q}$ in this convention. Time reversal gives the excitons at $\mathbf{Q}$ and $-\mathbf{Q}$ the same energies and parities, so nothing physical changes, but every formula that pairs an electron and a hole must use the stored pairing (Sec. 14.4).

Archives. An archive `GI_ExcPh_QXXXX.npz` stores, for one exciton momentum **Q**, the complex amplitudes $G[\mathbf{q}, \nu, \lambda, \beta] = \langle \beta, \mathbf{Q} + \mathbf{q} | \partial V_\nu | \lambda, \mathbf{Q} \rangle = \mathcal{G}_{\beta\lambda\nu}(\mathbf{Q}, \mathbf{q})$ in eV, for every $\mathbf{q}$ of the source mesh, every branch ν , initial state λ at **Q** and final state β at $\mathbf{Q} + \mathbf{q}$, together with $|\mathcal{G}|^2$, its electron and hole parts, and the energies. The final envelope is the conjugated one, as in YamboPy [6].

13. Exciton-phonon matrix elements

13.1 Definition

The exciton-phonon matrix element is the sum of a hole and an electron term, in each of which one carrier scatters while the other is a spectator [7, 8]:

$$\begin{aligned} \mathcal{G}_{\beta\lambda\nu}(\mathbf{Q}, \mathbf{q}) = & \sum_{v,v',c,c',\mathbf{k}} A_{\lambda,\mathbf{Q}}^{v,c,\mathbf{k}} [g_{vv',\nu}(\mathbf{k} - \mathbf{Q}, \mathbf{q}) \delta_{c,c'}] \left(A_{\beta,\mathbf{Q}+\mathbf{q}}^{v',c',\mathbf{k}} \right)^* \\ & - \sum_{v,v',c,c',\mathbf{k}} A_{\lambda,\mathbf{Q}}^{v,c,\mathbf{k}} [g_{c'c,\nu}^*(\mathbf{k} + \mathbf{q}, \mathbf{q}) \delta_{v,v'}] \left(A_{\beta,\mathbf{Q}+\mathbf{q}}^{v',c',\mathbf{k}+\mathbf{q}} \right)^*. \end{aligned} \quad (1)$$

13.2 Gauge

Both factors in Eq. (1) are built from the same Kohn-Sham wavefunctions (one Yambo SAVE for LetzElPhC and the BSE), so they share one single-particle gauge. Two freedoms remain on the exciton side.

Overall phases. Replacing $A^\lambda \rightarrow e^{i\theta_\lambda} A^\lambda$ and $A^\beta \rightarrow e^{i\theta_\beta} A^\beta$ multiplies $\mathcal{G}_{\beta\lambda\nu}$ by $e^{i(\theta_\lambda - \theta_\beta)}$, which cancels in $|\mathcal{G}_{\beta\lambda\nu}|^2$.

Degenerate manifolds. Within a degenerate manifold $\mathcal{M}$ the solver's basis is arbitrary: $|\beta'\rangle = \sum_{\beta \in \mathcal{M}} U_{\beta\beta'} |\beta\rangle$ with U unitary. Then $\sum_{\beta' \in \mathcal{M}} |\mathcal{G}_{\beta'\lambda\nu}|^2 = \sum_{\beta \in \mathcal{M}} |\mathcal{G}_{\beta\lambda\nu}|^2$, and likewise for the initial manifold. Only sums over complete manifolds are basis-independent, and every invariant used below is such a sum.

13.3 Excitons on the full zone

The BSE is solved only at the irreducible momenta (61 for a 24×24 mesh, 19 for 12×12). The exciton at any other momentum is the image of its irreducible parent under the operation R that maps one to the other [9]:

$$A_{vc,R\mathbf{k}}^{S,R\mathbf{Q}} = \sum_{v'c'} D_{cc'}^c(R; \mathbf{k}) A_{v'c',\mathbf{k}}^{S\mathbf{Q}} [D_{vv'}^v(R; \mathbf{k} - \mathbf{Q})]^*, \quad (2)$$

where $D^n(R; \mathbf{k})$ are the single-particle rotation matrices in the BSE band window, computed by LetzElPhC. For an operation that includes time reversal the parent envelope is complex conjugated first. The point is not to fix phases – they cancel anyway – but to make every member of a star the exact symmetry image of the same parent; a mapping that does not do so breaks the symmetry of $|\mathcal{G}|^2$.

13.4 Symmetry tests

Star uniformity. The integrated strength $T(\mathbf{q}) = \sum_{\nu\lambda\beta} |\mathcal{G}_{\beta\lambda\nu}(\mathbf{Q} = 0, \mathbf{q})|^2$, summed over all branches and complete manifolds, must be identical for every member of a star [10–12]. For GaN the ratio $T_{\max}/T_{\min}$ within a star is 1.0001 for the reconstructed full zone, against 2.25 for an initial output whose excitons at the star members were not constructed as images of a common parent.

Manifold-resolved test (xphd check-archive). For an archive at a time-reversal-invariant $\mathbf{Q}$, define for each $\mathbf{q}$ $S_{abc}(\mathbf{q}) = \sum_{\nu \in a} \sum_{\lambda \in b} \sum_{\beta \in c} |\mathcal{G}_{\beta\lambda\nu}(\mathbf{Q}, \mathbf{q})|^2$ over phonon manifolds a (branches within 0.1 meV), initial manifolds b and final manifolds c (states within a tolerance, default 15 meV). Symmetry requires $S(\mathbf{q}) = S(g\mathbf{q})$ for g = time reversal ($\mathbf{q} \rightarrow -\mathbf{q}$) and $g = C_3$. The reported deviation is

$$L_1 = \frac{\sum_{\mathbf{q}} \sum_{abc} |S_{abc}(\mathbf{q}) - S_{abc}(g\mathbf{q})|}{\frac{1}{2} \sum_{\mathbf{q}} \sum_{abc} [S_{abc}(\mathbf{q}) + S_{abc}(g\mathbf{q})]}, \quad (3)$$

with OK below 10^{-2} and FAIL above 5×10^{-2} . The C_3 matrix in reduced coordinates depends on the angle between $\mathbf{b}_1$ and $\mathbf{b}_2$: $(q_1, q_2) \rightarrow (-q_1 - q_2, q_1)$ for 60° and $(-q_2, q_1 - q_2)$ for 120° ; the code takes the one that leaves the exciton energies invariant. Results: GaN 1.3×10^{-3} (time reversal) and 3.3×10^{-3} (C_3); WSe₂ 1.2×10^{-3} and 1.0×10^{-3} . A rotation fault shows as one operation far above the other.

Exchange-split doublets. At $\mathbf{Q} = 0$ the BSE evaluates the long-range exchange along one direction $\hat{\mathbf{e}}$, splitting each E' doublet into members polarised parallel and perpendicular to $\hat{\mathbf{e}}$ (GaN: 2.8 meV for the first bright doublet, 10 meV for the second). For a member with real in-plane dipole $\mathbf{d}$, the coupling to small- $\mathbf{q}$ polar phonons carries the factor $|\hat{\mathbf{q}} \cdot \mathbf{d}|^2$. It is unchanged by $\mathbf{q} \rightarrow -\mathbf{q}$ but not by C_3 , which rotates $\hat{\mathbf{q}}$ relative to the fixed $\mathbf{d}$. Summed over the doublet, $|\hat{\mathbf{q}} \cdot \mathbf{d}_{\parallel}|^2 + |\hat{\mathbf{q}} \cdot \mathbf{d}_{\perp}|^2 = |\mathbf{d}|^2$ is isotropic. With 5 meV manifolds the GaN second doublet is split in two and the C_3 test fails ($L_1 = 0.20$, all of it in that doublet and the LO branch at the smallest $|\mathbf{q}|$); grouping it restores 3.3×10^{-3} .

14. Mirror parity and selection rules

14.1 The horizontal mirror

σ_h maps $(x, y, z) \rightarrow (x, y, -z)$ and leaves every in-plane wavevector unchanged, so it belongs to the little group of every $\mathbf{Q}$ and commutes with the Kohn-Sham, phonon and BSE Hamiltonians at every point of the zone. Every non-degenerate state therefore carries an exact parity across the whole zone. In a planar layer every atom lies in the mirror plane and maps onto itself; in 1H-WSe₂ the mirror maps W onto itself and swaps the two Se, $\pi = [0, 2, 1]$.

14.2 Electrons

For spinless states the diagonal of $D(\sigma_h; \mathbf{k})$ gives $p_n(\mathbf{k}) = \langle \psi_{n\mathbf{k}} | \sigma_h | \psi_{n\mathbf{k}} \rangle = \pm 1$. With spin-orbit coupling $\sigma_h = I C_{2z}$ acts on spin as $e^{-i\pi\sigma_z/2} = -i\sigma_z$, so $\sigma_h^2 = -1$ and the eigenvalues are $\pm i$. The label is $\lambda_n = \text{Im } p_n = \pm 1$, chosen so that an exciton's parity is still a product of labels (Sec. 14.4). WSe₂ gives $\langle D(\sigma_h)^2 \rangle = -1.0000$, confirming that the spin rotation is included. At time-reversal-invariant points (Γ , M) a spinor band is a Kramers pair with opposite eigenvalues, so no single label exists there. Time reversal maps $\mathbf{k} \rightarrow -\mathbf{k}$ and $+i \rightarrow -i$, so each band's label flips between $\mathbf{k}$ and $-\mathbf{k}$ and every band is exactly 50% even: at the $\mathbf{K}$ band edges, where the orbitals are mirror-even d states, the label is the out-of-plane spin, and this is spin-valley locking written as mirror parity.

14.3 Phonons

σ_h acts on a displacement pattern $\mathbf{e} = (\mathbf{e}_{\kappa})$ as $(\sigma_h \mathbf{e})_{\kappa} = M \mathbf{e}_{\pi(\kappa)}$ with $M = \text{diag}(1, 1, -1)$ and π the atom permutation (an involution). The parity of a normalised mode is

$$p_{\nu\mathbf{q}} = \langle \mathbf{e}_{\nu\mathbf{q}} | \sigma_h \mathbf{e}_{\nu\mathbf{q}} \rangle = \sum_{\kappa} \mathbf{e}_{\nu\mathbf{q},\kappa}^{\dagger} M \mathbf{e}_{\nu\mathbf{q},\pi(\kappa)} = \pm 1. \quad (4)$$

For a planar layer, $\pi = \text{id}$ and $p = 1 - 2w_z$ with $w_z = \sum_{\kappa} |e_{\kappa z}|^2$: out-of-plane modes are odd. In WSe₂ this shortcut fails for 34.4% of $(\mathbf{q}, \nu)$: the Se breathing mode A'_1 moves along z yet is even, and the Se antiphase mode E'' moves in plane

yet is odd. Equation (4) gives ± 1 to 7×10^{-16} there. The eigenvectors are taken from the electron-phonon database itself, in the order of the couplings: branches labelled from a separate `matdyn.x` run, ordered by its own frequencies, disagreed at 5% of the points in GaN.

14.4 Excitons

Applying a rotation R to exciton T and projecting on S ,

$$D_{ST}(R) = \sum_{\mathbf{k}} \sum_{vc, v'c'} [A_{vc\mathbf{k}}^S]^* D_{cc'}^c(R; \mathbf{k}) A_{v'c', R^{-1}\mathbf{k}}^T [D_{vv'}^v(R; \mathbf{k} - \mathbf{Q})]^*. \quad (5)$$

For σ_h , $R^{-1}\mathbf{k} = \mathbf{k}$, and for non-degenerate bands

$$\chi_S \equiv D_{SS}(\sigma_h) = \sum_{vc\mathbf{k}} |A_{vc\mathbf{k}}^S|^2 p_c(\mathbf{k}) p_v^*(\mathbf{k} - \mathbf{Q}), \quad (6)$$

real and equal to ± 1 for spinless and spinor states alike (for spinors $p_c p_v^* = (\pm i)(\mp i)$ or $(\pm i)(\pm i)^*$). Using Yambo's storage, the valence matrix must be taken at the hole's wavevector, $\mathbf{k} + \mathbf{Q}$ in table indices. Of the four ways of attaching the shift, only this one gives $|\chi_S| = 1$ for every state (100% of states at all 61 irreducible $\mathbf{Q}$ in GaN, against at most 35% for the others); the wrong pairing produced a fictitious ring of fractional parity around Γ .

14.5 Selection rule

The phonon perturbation transforms like its mode, $\sigma_h \partial V_\nu \sigma_h^{-1} = p_\nu \partial V_\nu$. Inserting $\sigma_h^\dagger \sigma_h = 1$ twice,

$$\mathcal{G}_{\beta\lambda\nu} = \langle \beta | \sigma_h^\dagger (\sigma_h \partial V_\nu \sigma_h^{-1}) \sigma_h | \lambda \rangle = p_\beta^* p_\nu p_\lambda \mathcal{G}_{\beta\lambda\nu}, \quad (7)$$

so $\mathcal{G}_{\beta\lambda\nu} \neq 0$ only if $p_\beta p_\nu p_\lambda = +1$: an even phonon connects excitons of equal parity, an odd phonon excitons of opposite parity.

Quantifying it (xphd selection-rule). Each term $|\mathcal{G}_{\beta\lambda\nu}(\mathbf{q})|^2$ is classed as allowed, forbidden or undefined (a final state with $|\chi_\beta|$ below a tolerance). Results:

- GaN, bright state at Γ : forbidden/allowed = 5×10^{-7} .
- WSe₂, dark (odd) and bright (even) states at Γ : 1.4×10^{-4} and 1.0×10^{-3} at tolerance 0.5, carried entirely by one pair of final states 0.13 meV apart and of opposite parity, at one star of 12 $\mathbf{q}$, returned mixed by the solver ($|\chi| = 0.915, 0.951$). At tolerance 0.96: 1.8×10^{-7} and 1.4×10^{-7} , with 22% and 49% of the coupling then undefined. With manifold rotation (Sec. 15), state 1: 6.3×10^{-6} with 5.9% undefined.

`where_forbidden.py` separates three fingerprints of residual forbidden weight: a phonon swap (labels and couplings in different orders), a state mix (a near-degenerate opposite-parity partner, or $|\chi| < 0.95$), and a diffuse admixture $\varepsilon = (1 - |\chi|)/2$ (forbidden/allowed into a state scales as ε , state by state).

14.6 Irreducible representations at $\mathbf{K}$ and the emission rule

The little group at $\mathbf{K}$ is C_{3h} , Abelian, with six one-dimensional irreducible representations (Table 1). Time reversal maps $\mathbf{K}$ to $\mathbf{K}'$ and adds no degeneracy at $\mathbf{K}$; $\sigma_v T$ fixes $\mathbf{K}$ but maps each C_3 eigenvalue to itself and squares to $+1$, so it adds none either. Which of A, E_1, E_2 labels a state depends on the C_3 axis (through Ga, N or the hexagon centre), because a Bloch sum at $\mathbf{K}$ acquires a phase from sites off the axis; the σ_h parity does not.

Table 1: Character table of C_{3h} , $\omega = e^{2\pi i/3}$.

	E	C_3	C_3^2	σ_h	S_3	S_3^5
A'	1	1	1	1	1	1
A''	1	1	1	-1	-1	-1
E'_1	1	ω	ω^2	1	ω	ω^2
E'_2	1	ω^2	ω	1	ω^2	ω
E''_1	1	ω	ω^2	-1	$-\omega$	$-\omega^2$
E''_2	1	ω^2	ω	-1	$-\omega^2$	$-\omega$

For the dark A'' exciton at $\mathbf{K}$ to emit, a phonon must carry away the momentum $\mathbf{K}$ while a photon is emitted. The exciton must be contained in the product of the phonon and the photon,

$$\Gamma_{\text{exc}} \subset \Gamma_\nu \otimes \Gamma_{\text{photon}}, \quad (8)$$

a form invariant under a change of the C_3 axis (the conjugated form $\Gamma_\nu \subset \Gamma_{\text{exc}}^* \otimes \Gamma_{\text{photon}}$ renames the helicities but selects the same branch). In-plane light is E' in D_{3h} and splits at $\mathbf{K}$ into $E'_1 \oplus E'_2$, one component per helicity. For A'' , a

Table 2: GaN phonons at **K** (C_3 about the [Ga/N] site) and whether each can mediate the emission of the A'' exciton under Eq. (8).

Branch	$\hbar\omega$ (meV)	ω (cm^{-1})	Irrep	σ_h	Mediates
1	16.2	130.8	E_1''	odd	yes
2	17.9	145.1	A''	odd	no
3	28.7	231.4	E_2'	even	no
4	42.4	341.8	A'	even	no
5	93.7	756.3	E_1'	even	no
6	100.4	810.3	E_2'	even	no

photon of E_1' symmetry requires an E_2'' phonon and one of E_2' symmetry an E_1'' phonon. Only E_1'' occurs among the six GaN branches at **K** (Table 2), so one branch mediates the emission and its light has a single helicity; time reversal gives **K'** the opposite one.

When is the sideband fully polarised? Let the C_3 eigenvalues of the two flexural modes at **K** be ϕ_1, ϕ_2 (two of $1, \omega, \omega^2$) and that of the dark exciton ε . By Eq. (8) a photon of eigenvalue ω needs a phonon of eigenvalue $\varepsilon\omega^2$, one of eigenvalue ω^2 a phonon of $\varepsilon\omega$. Both helicities have a partner only if $\{\varepsilon\omega, \varepsilon\omega^2\} = \{\phi_1, \phi_2\}$, i.e. only if ε is the third eigenvalue. If ε equals ϕ_1 or ϕ_2 , one helicity is available and each valley's sideband is fully circularly polarised. Shifting the axis multiplies $\varepsilon, \phi_1, \phi_2$ by the same phase, so the criterion is axis-independent. In GaN the two flexural modes at **K** are sublattice-polarised (each moves one sublattice), E_1'' at 16.2 meV and A'' at 17.9 meV; the A'' exciton ($\varepsilon = 1$) coincides with the A'' mode, so only E_1'' mediates.

14.7 Two classes of planar D_{3h} monolayers

In a planar honeycomb binary each band edge is π -like (p_z , odd) or σ -like ($s, p_{x,y}$, even), and for an exciton dominated by one transition $\chi = p_c p_v$. The valence maximum is the anion p_z state at **K** (odd); the class is set by the conduction minimum (Table 3). *Class I*: both edges π -like at **K**; the lowest exciton is the bright, even E' state at Γ and its sidebands are in-plane optical replicas ($E' \subset E' \otimes E'$; the flexural branch is forbidden). *Class II*: a σ -like conduction band at Γ lies lowest; the lowest exciton is momentum-dark and odd at **K**, on a band disconnected from the bright one, and its sideband needs an odd phonon at **K**.

Table 3: The two classes. Rows marked await the fixed-code runs.

	VBM	CBM	lowest exciton	sideband phonon	$ P $
hBN (I)	K , N p_z , odd	K , B p_z , odd	Γ , even, E' (5.475 eV)	TO/LO, even	[0]
SiC (I)	K , C p_z , odd	K , Si p_z , odd	Γ , even, E'	[TO/LO]	[0]
GaN (II)	K , N p_z , odd	Γ , Ga s , even	K , odd, A'' (3.429 eV)	E_1'' , odd, 16.2 meV	1.000

15. Near-degenerate manifolds

15.1 Why near-degenerate states mix

Let two states of opposite parity be separated by Δ , and let the computed BSE Hamiltonian break σ_h by a small δ between them (numerical noise). First-order mixing gives an admixture amplitude δ/Δ , i.e. $|\chi| = 1 - 2(\delta/\Delta)^2$. For $\Delta = 0.13$ meV, $\delta \sim 10$ μeV already gives $|\chi| \approx 0.99$, and the WSe_2 pair came back at 0.915 and 0.951. Per-state parity then either counts a forbidden residue or excludes the states.

15.2 Splitting by the manifold's parity eigenstates

Group the states at each **Q** into manifolds of energies closer than a tolerance (xphd parity -manifold-tol, default 5 meV) and compute, with Eq. (5), the matrix $M_{mn} = \langle m | \sigma_h | n \rangle$ within each. For a closed manifold M is unitary and Hermitian, with eigenvalues ± 1 ; its eigenvectors v_i are the parity eigenstates the solver could equally have returned. The state $\partial V_\nu | \lambda \rangle$ projected on the manifold is $|\psi\rangle = \sum_n G_n | n \rangle$ with $G_n = \langle n | \partial V_\nu | \lambda \rangle$, the archive's amplitudes. Its weight on eigenvector i is

$$W_i = |\langle v_i | \psi \rangle|^2 = \left| \sum_n (v_i)_n^* G_n \right|^2, \quad \sum_i W_i = \sum_n |G_n|^2, \quad (9)$$

and W_i is allowed or forbidden according to the eigen-parity $\mu_i = \pm 1$. Nothing is excluded for being mixed; only eigenvectors with $|\mu_i|$ below the tolerance remain undefined. The weights in each parity sector, $\langle \psi | \frac{1}{2}(1 \pm M) | \psi \rangle$, do not depend on the basis within the manifold.

Rotated states. An archive's final state at $\mathbf{Q}+\mathbf{q}$ is the image of its irreducible parent, Eq. (2). For spatial R , $R^{-1}\sigma_h R = \sigma_h$ on excitons, so the manifold matrix is the parent's; for an operation containing time reversal (antiunitary) it is M^* . (For spinors, σ_h anticommutes with C'_2 and σ_v in the double group; the electron and hole each acquire the sign and it cancels for the exciton, which is why per-state exciton parity is constant over a star.) The time-reversal flag per full-zone point is the one xphd generate uses: symmetry index $\geq n_{\text{sym}}/(1+t_{\text{rev}})$.

Result. In WSe_2 (state 1) the rotation lowers forbidden/allowed from 1.4×10^{-4} to 6.3×10^{-6} without excluding anything beyond the 5.9% already undefined. Only 70% of the eigen-parities lie within 0.02 of ± 1 .

15.3 Closure diagnostics

An eigen-parity short of ± 1 means the manifold is not closed: its partner is missing.

- *Cut by the state count* (manifold_closure.py): compare the eigen-parities of each manifold cut at the archive's state count with those of the complete manifold from a parity run with more states. In WSe_2 , 75.8% within 0.02 either way: the cut is not the cause.
- *Partner beyond the tolerance*: widening the manifolds from 5 to 20 meV changed nothing (75.3% to 75.0%).
- *Window not closed* (window_closure.py): σ_h maps a band partly outside the BSE window when its mirror partner lies outside it. Then the row norm $\sum_{j \in \text{window}} |D_{ij}(\sigma_h; \mathbf{k})|^2 < 1$, and every exciton built on it maps partly outside the BSE space. [Run on WSe_2 .]

16. Exciton-phonon linewidths

16.1 Golden rule

To first order in the coupling and in the dilute limit, the full width at half maximum of state λ at $\mathbf{Q}$ is

$$\Gamma_{\lambda\mathbf{Q}}(T) = 2\pi \sum_{\beta\nu} \int_{\text{BZ}} \frac{A_{\text{uc}} d^2q}{(2\pi)^2} |\mathcal{G}_{\beta\lambda\nu}(\mathbf{Q}, \mathbf{q})|^2 \left[(N_{\nu\mathbf{q}} + 1) \delta(E_{\lambda\mathbf{Q}} - E_{\beta\mathbf{Q}+\mathbf{q}} - \hbar\omega_{\nu\mathbf{q}}) + N_{\nu\mathbf{q}} \delta(E_{\lambda\mathbf{Q}} - E_{\beta\mathbf{Q}+\mathbf{q}} + \hbar\omega_{\nu\mathbf{q}}) \right], \quad (10)$$

the imaginary part of the Fan-Migdal self-energy, with the final-state exciton occupation set to zero. Since the zone has area $(2\pi)^2/A_{\text{uc}}$, the measure is $A_{\text{uc}} d^2q/(2\pi)^2 = d^2\tilde{q}/\tilde{A}_{\text{BZ}} = (2/\sqrt{3}) d^2\tilde{q}$ for $\tilde{q}$ in units of $|\mathbf{b}|$, and $(1/N_q) \sum_{\mathbf{q}}$ on a mesh. States within 2 meV are one manifold, their widths averaged; the 2.8 meV exchange-split bright doublet of GaN is reported state by state.

16.2 Linear-triangle integration

On the 24×24 mesh a Lorentzian or Gaussian δ would need a width large enough to make the result depend on it, and in two dimensions it blurs the kinematic thresholds. The δ functions are integrated analytically [13, 14]: each mesh cell is split into two triangles, and in each the energy difference $f = E_{\lambda} - E_{\beta} \mp \hbar\omega$ is linear between its vertex values $f_1 \leq f_2 \leq f_3$. The values of a linear function over a triangle of area A_T are distributed with density

$$\rho_T(\varepsilon) = \frac{1}{A_T} \int_T \delta(\varepsilon - f) d^2r = \begin{cases} \frac{2(\varepsilon - f_1)}{(f_2 - f_1)(f_3 - f_1)}, & f_1 \leq \varepsilon < f_2, \\ \frac{2(f_3 - \varepsilon)}{(f_3 - f_1)(f_3 - f_2)}, & f_2 \leq \varepsilon < f_3, \\ 0 & \text{otherwise,} \end{cases} \quad (11)$$

normalised to $\int \rho_T d\varepsilon = 1$. The integral of Eq. (10) is then $\sum_T w_T A_T \rho_T(0)$, with w_T the (piecewise-constant) weight $|\mathcal{G}|^2(N+1)$ or $|\mathcal{G}|^2N$ of that triangle. This is done on a mesh refined 15-fold, to 360×360 (factor 30 for WSe_2), which needs the phonon and exciton energies there.

16.3 Inputs on the refined mesh

Phonons. Diagonalising the DFPT dynamical matrices directly with `matdyn.x`, including the two-dimensional non-analytic correction that removes the spurious LO-TO splitting at Γ [15]. They reproduce the source mesh to 1.7 μeV on all branches, apart from a 0.237 meV acoustic residue at Γ removed by the acoustic sum rule.

Exciton energies. Exciton bands are indexed by ascending energy, so where two bands cross each indexed band has a kink. A global Fourier interpolant [16, 17] rings from every kink across the zone; at the $\mathbf{K}$ minimum of GaN it made the band 34% too steep and gave it a spurious twofold anisotropy. Each refined energy is instead taken from a quadratic about the nearest BSE point,

$$E(u, v) = E_0 + g_1 u + g_2 v + A u^2 + B u v + D v^2, \quad (12)$$

with E_0 the BSE energy and (u, v) the reduced offset. The six neighbours lie at $\pm(h, 0)$, $\pm(0, h)$, $\pm(h, -h)$. Sums over opposite pairs cancel the odd terms and fix the curvature exactly:

$$S_1 \equiv E(h, 0) + E(-h, 0) - 2E_0 = 2A h^2, \quad S_2 = 2D h^2, \quad S_3 = 2(A - B + D) h^2, \quad (13)$$

so $A = S_1/2h^2$, $D = S_2/2h^2$, $B = (S_1 + S_2 - S_3)/2h^2$. Differences give three estimates of the gradient, $d_1 = E(h, 0) - E(-h, 0) = 2hg_1$, $d_2 = 2hg_2$ and $d_3 = E(h, -h) - E(-h, h) = 2h(g_1 - g_2)$, each up to third-order terms. Minimising $(d_1 - x)^2 + (d_2 - y)^2 + (d_3 - x + y)^2$ over $x = 2hg_1$, $y = 2hg_2$ gives

$$g_1 = \frac{2d_1 + d_2 + d_3}{6h}, \quad g_2 = \frac{d_1 + 2d_2 - d_3}{6h}. \quad (14)$$

The residual is the odd, trigonal-warping part of the band, third order in the offset. The quadratic reproduces E_0 exactly and its neighbours to third order, and because it uses only nearest neighbours a crossing affects only the cells next to it. Quadratic interpolation onto a refined mesh followed by linear-triangle integration is the hybrid tetrahedron scheme of MacDonald *et al.* [18, 19], here in two dimensions [20]. We centre each quadratic on a mesh point rather than fitting one per triangle through its vertices and edge midpoints: such per-triangle quadratics are continuous but kinked at the vertices, and a band minimum always lies on a vertex.

Tests. On model bands with closed-form linewidths, the per-triangle scheme erred by up to 17% at a minimum and the present one by at most 2.4%; a crossing 150 meV above the minimum shifted the Fourier result by 35% and the present one by 2%. At the GaN **K** minimum, fitting the curvature to the first, second and third shells of BSE points gives 12.44, 12.26 and 12.18 eV $|b|^{-2}$. Predicting every other BSE point from the rest gives an rms error of [X] meV for this scheme and [Y] meV for Fourier interpolation (`compare_band_interp.py`). A Wannier representation of the exciton [21] would treat crossings rigorously.

Couplings. $|G|^2$ is constant within each coarse cell, except for the three acoustic branches in the cell around $\mathbf{q} = 0$, where the coupling vanishes at the cell's own mesh point. There we use the long-wavelength form $|G|^2 2\hbar\omega = Cq^2$ (i.e. $|G|^2 \propto q$ for a linear branch), with C fitted for each pair of initial and final states to the innermost shell of the source mesh. Phonons below 0.5 meV are excluded. In `xphd linewidth` this is `-acoustic-model q2 -n-shell 1`; the default is `none`.

Symmetry. Each state's width is invariant under the point group, so Eq. (10) is evaluated at the irreducible **Q** (61 for 24×24 , D_{3h} with time reversal) and unfolded.

16.4 The band minimum in closed form

At the global minimum (GaN: **K**) no lower state exists; the width comes only from absorption of acoustic phonons within the valley, into final states inside the first mesh cell, where band and coupling are extrapolated. Model the valley as $E(q) = E_{\min} + bq^2$, the branch as $\hbar\omega = \hbar cq$ and the coupling as $|G|^2 2\hbar\omega = Cq^2$, with q in units of $|b|$, following the deformation-potential treatment of exciton-acoustic-phonon scattering [22–24]. Absorption from the bottom conserves energy when $E_{\min} + \hbar cq = E_{\min} + bq^2$, i.e. at

$$q^* = \frac{\hbar c}{b}, \quad \hbar\omega^* = \hbar cq^* = \frac{(\hbar c)^2}{b} \quad (= 2Mc^2 \text{ for } b = \hbar^2/2M). \quad (15)$$

Then, with the measure of Sec. 16.1 and $d^2q = 2\pi q dq$,

$$\begin{aligned} \Gamma &= 2\pi \cdot \frac{2}{\sqrt{3}} \int 2\pi q dq \frac{Cq}{2\hbar c} N(\hbar cq) \delta(bq^2 - \hbar cq) = \frac{8\pi^2}{\sqrt{3}} \frac{Cq^{*2}}{2\hbar c} \frac{N(\hbar\omega^*)}{|2bq^* - \hbar c|} \\ &= \frac{4\pi^2}{\sqrt{3}} \frac{Cq^{*2}}{(\hbar c)^2} N(\hbar\omega^*) = \frac{4\pi^2}{\sqrt{3}} \frac{C N(\hbar\omega^*)}{b^2}, \end{aligned} \quad (16)$$

using $|2bq^* - \hbar c| = \hbar c$. Above $\hbar\omega^*/k_B$, $N \approx k_B T/\hbar\omega^*$ and $\Gamma \approx (4\pi^2/\sqrt{3}) C k_B T/[b(\hbar c)^2]$ grows linearly in T ; below it the width freezes out. For GaN all inputs come from our data: $b = 12.44$ eV $|b|^{-2}$, $c_{LA} \approx [9.1]$ km/s, $\hbar\omega^* = 1.49$ meV ($q^* = 0.011 |b|$), and C from the minimum-to-minimum coupling (implying a deformation potential $\Xi \approx [8.4]$ eV; TA gives 10^{-3} of LA and ZA nothing, by parity). Equation (16) gives 0.09, 10.6–11.4 and 45–48 meV at 5, 77 and 300 K (the range from extrapolating C to $q \rightarrow 0$); the code gives 0.087, 10.4 and 50.5 meV, the excess at 300 K being about 6 meV of absorption beyond the first cell. At 77 K the width exceeds $\hbar\omega^*$: the scattering is quasi-elastic and the line not strictly Lorentzian.

Leaving the state versus leaving the valley. The absorbed phonon moves the exciton by $|q^*| = 0.011 |b|$ inside its valley, with almost no change of energy. Excluding the first cell (`-acoustic-model none`) leaves the rate of scattering beyond it, the escape from the valley bottom: 0.044 meV at 77 K ($\hbar/\Gamma \approx 15$ ps) and [X] meV at 300 K.

16.5 Off-diagonal self-energy and validity

The off-diagonal elements of the exciton-phonon self-energy [25] change the absorption lineshape by less than [0.3] %: the diagonal width of the lowest bright exciton (51.9 meV at 77 K) exceeds the largest off-diagonal element ([2.4] meV) by a factor [22]. For the highest states, widths reach several hundred meV (lifetimes ~ 1 fs, comparable to the band spacing); there Eq. (10) no longer describes a sharp resonance.

GaN values. Bright state at Γ : 36.7, 51.9 and 150.9 meV at 5, 77 and 300 K (12.7 fs at 77 K, 4.4 fs at 300 K; 99.8% emission at 5 K, 43% absorption at 300 K); its partner, 56.8 meV at 77 K. **K** minimum: 0.087, 10.4 (63 fs) and 50.5 meV (13 fs).

17. Phonon-assisted luminescence and helicity

17.1 Second-order emission

An exciton β at momentum $\mathbf{q}$ emits through a virtual bright state λ at $\mathbf{Q} = 0$, emitting or absorbing a phonon $(\mu, \mathbf{q})$ that carries the momentum [26, 27]:

$$I^{\text{PL}}(\omega; T) \propto \frac{1}{N_q} \sum_{\beta, \mathbf{q}, \mu, \pm} \left| \sum_{\lambda} \frac{T_{\lambda} \mathcal{G}_{\lambda\beta\mu}(\mathbf{q}, -\mathbf{q})}{E_{\beta\mathbf{q}} \mp \hbar\omega_{\mu\mathbf{q}} - E_{\lambda 0}} \right|^2 \times N_{\beta\mathbf{q}}(T_{\text{exc}}) F_{\mu\mathbf{q}}^{\pm}(T) \delta(\hbar\omega - E_{\beta\mathbf{q}} \pm \hbar\omega_{\mu\mathbf{q}}), \quad (17)$$

with T_{λ} the dipole of the bright state, $N_{\beta\mathbf{q}} \propto e^{-E_{\beta\mathbf{q}}/k_B T_{\text{exc}}}$ and $F^{\pm} = N_{\mu\mathbf{q}} + \frac{1}{2} \pm \frac{1}{2}$. In GaN ($T = T_{\text{exc}} = 10$ K, 15 states, six branches, 5 meV broadening) the spectrum below the direct line is a single satellite at 3.4129 eV, 16.1 meV below the dark $\mathbf{K}$ exciton (3.4290 eV), matching the E_1'' branch at $\mathbf{K}$ (16.2 meV); branch-resolved, it exceeds every other branch by seven orders of magnitude.

17.2 Helicity and valley

Replace T_{λ} by $\hat{\epsilon}_{\pm}^* \cdot \mathbf{d}_{\lambda}$ with $\hat{\epsilon}_{\pm} = (\hat{x} \pm i\hat{y})/\sqrt{2}$ and the complex Cartesian dipoles $\mathbf{d}_{\lambda}$ of the $\mathbf{Q} = 0$ excitons, and separate the emitters β by valley. The degree of circular polarisation is $P = (I_+ - I_-)/(I_+ + I_-)$. Because $\hat{\epsilon}_{\pm}$ form an orthonormal basis of the plane, for any amplitude vector $\mathbf{v}$,

$$|\hat{\epsilon}_+^* \cdot \mathbf{v}|^2 + |\hat{\epsilon}_-^* \cdot \mathbf{v}|^2 = |v_x|^2 + |v_y|^2, \quad (18)$$

so $I_+ + I_-$ equals the helicity-summed spectrum exactly (a sum rule checked to 3.7×10^{-16} in GaN). Results: $P = -1.0000$ from $\mathbf{K}$, $+1.0000$ from $\mathbf{K}'$, equal valley intensities to 0.12% (ratio 0.9988). The dipoles must come from a single BSE solution: with dipoles and eigenvectors from different runs, or with valleys summed, the helicity is lost. In hBN (class I) no valley-locked satellite is expected [(negative control to run)].

17.3 Exciton dipoles and oscillator strengths

xphd dipoles builds $\mathbf{d}_{\lambda} = \sum_{v\mathbf{c}\mathbf{k}} A_{v\mathbf{c}\mathbf{k}}^{\lambda} \mathbf{d}_{c\mathbf{v}}(\mathbf{k})$ from the single-particle dipoles of `ndb.dipoles` in the BSE window, rotated to the full zone like the envelopes (complex conjugated for time-reversed points). It is validated against Yambo's oscillator strengths manifold by manifold: the ratio is the same constant (the unit conversion) for every manifold, with exchange-split pairs compared as sums. For the transport, `add_osc.py` writes the in-plane strengths $|d_x|^2 + |d_y|^2$ into the Γ archive (field `osc_Q0`); σ_h -odd, z -polarised excitons receive none.

18. Real-time exciton Boltzmann transport

18.1 Equation and rates

For the occupations $F_i(t)$, $i \equiv (\lambda, \mathbf{Q})$ [8, 28],

$$\frac{dF_i}{dt} = - \sum_j \left[P_{ij} F_i (1 + F_j) - P_{ji} F_j (1 + F_i) \right], \quad (19)$$

$$P_{ij} = \frac{2\pi}{\hbar} \frac{1}{N_q} \sum_{\nu} |\mathcal{G}_{\beta\lambda\nu}(\mathbf{Q}, \mathbf{q})|^2 \left[(N_{\nu\mathbf{q}} + 1) \delta(E_i - E_j - \hbar\omega_{\nu\mathbf{q}}) + N_{\nu\mathbf{q}} \delta(E_i - E_j + \hbar\omega_{\nu\mathbf{q}}) \right], \quad (20)$$

with $\mathbf{q} = \mathbf{Q}_j - \mathbf{Q}_i$. With the matrix elements, phonons and triangle weights of Sec. 16, the row sums are the linewidths, $\sum_j P_{ij} = \Gamma_i/\hbar$; they agree to better than 0.1% with Eq. (10) evaluated on the same mesh (without the in-cell acoustic channel, which moves an exciton within its own cell and cannot move population between mesh states). The rate matrix couples different momenta, so it is assembled on the full zone: $576 \times 5 = 2880$ states for GaN.

18.2 Detailed balance

A transition $i \rightarrow j$ by emission is integrated over the triangle containing $\mathbf{q}$, the reverse $j \rightarrow i$ by absorption over the one containing $-\mathbf{q}$. The two energy-conserving surfaces are identical by time reversal, but linear interpolation gives them different areas $A_{ij}^{\text{em}} \neq A_{ji}^{\text{abs}}$, and the system then accumulates particles and never equilibrates. With the symmetrised weight $B_{ij} = \frac{1}{2}(A_{ij}^{\text{em}} + A_{ji}^{\text{abs}})$,

$$P_{ij} = B_{ij} |\mathcal{G}|^2 (N + 1), \quad P_{ji} = B_{ij} |\mathcal{G}|^2 N \Rightarrow \frac{P_{ij}}{P_{ji}} = \frac{N + 1}{N} = e^{\hbar\omega/k_B T} = e^{(E_i - E_j)/k_B T} \quad (21)$$

exactly, for every energy-conserving pair. The rates are written without dividing by N , which would amplify weakly populated channels. Particle number is conserved to 10^{-9} .

Stationary state. With $F = 1/(e^{(E-\mu)/k_B T} - 1)$, $F/(1+F) = e^{-(E-\mu)/k_B T}$, so $P_{ij}F_i(1+F_j) = P_{ji}F_j(1+F_i)$ is exactly the ratio above: every pair is balanced, and the Bose-Einstein distribution is stationary (verified to 10^{-16}); at these densities it is Boltzmann.

18.3 Refining the rates

The full matrix cannot be refined like a linewidth (each refined $\mathbf{q}$ is a new state; the size grows as the fourth power of the refinement), but its row sums are ordinary zone integrals. With s_i the ratio of the row sum on a fivefold finer mesh to the coarse one, the matrix is rescaled as $P_{ij} \rightarrow P_{ij}\sqrt{s_i s_j}$, which leaves P_{ij}/P_{ji} unchanged. A one-sided rescaling $P'_{ij} = s_i P_{ij}$ would not: detailed balance $P'_{ij}F_i = P'_{ji}F_j$ then gives $F_i/F_j = (s_j/s_i)e^{-(E_i-E_j)/k_B T}$, i.e. $F_i \propto e^{-E_i/k_B T}/s_i$, the wrong equilibrium. Without the correction the coarse rates are low by **[2.5]**%; with it they match the refined reference to **[X]**%. The thermalisation time is set by the refined row sums; the branching ratios vary smoothly with $\mathbf{q}$.

18.4 Initial condition, integration, diagnostics

Optical injection fills $\mathbf{Q} = 0$: $F_\lambda(0) \propto |\mu_\lambda|^2 \exp[-(E_\lambda - E_{\text{pump}})^2/2\sigma^2]$ **[(pump as in the main text)]**, normalised to the injected density; results are unchanged from 10^{-4} to 10^{-2} per state (linear regime). Equation (19) is stiff and is integrated with an implicit BDF solver (relative and absolute tolerances 10^{-7} and 10^{-12}); occupations far above the minimum can dip to -10^{-15} , an integrator artefact. Two decompositions are reported: the filling rate $\sum_j P_{ji}^{(c)} F_j$ of a state by channel c , and the aggregate flux $\sum_{ij} F_i P_{ij}^{(c)}$; a channel can fill many sparsely populated states yet carry little flux.

18.5 Results and the finite-temperature equilibrium

GaN at 77 K: the population relaxes to 4.26 meV above the minimum, against a Boltzmann value of 4.09 meV on the same states. Both lie below $k_B T = 6.6$ meV, the value for a two-dimensional continuum, because the nearest mesh states around $\mathbf{K}$ lie $b\hbar^2 = 22$ meV above it. Of the population, 35, 50, 90 and 99% reaches the $\mathbf{K}$, $\mathbf{K}'$ valleys (within $0.12|\mathbf{b}|$) by 30, 41, 98 and 172 fs; it then escapes the valley bottom at 0.044 meV (15 ps), although each $\mathbf{K}$ state loses coherence in 63 fs. The valleys fill mainly through the fourth branch in its in-plane, LA-like form: valley and feeding states are both σ_h -odd, so the phonon must be even.

At a temperature where $k_B T$ is comparable to the energy of the first shell of mesh points around the minimum, the equilibrium on the mesh is not concentrated on the minimum's own point: a light exciton at 300 K can hold only $\sim 20\%$ there (`check_bte.py` prints the share at the minimum's point and in each valley against Boltzmann on the same states). A run has settled when it matches Boltzmann; equal emission and absorption flux for every branch is the same statement. The hBN run at 300 K behaves this way.

19. Quasiparticle corrections: decomposition and use

19.1 Newton solution and its parts

Yambo's linearised quasiparticle equation is

$$E = E_o + Z[\Sigma_x + \text{Re}\Sigma_c(E_o) - V_{xc}], \quad Z = [1 - \partial_\omega \text{Re}\Sigma_c|_{E_o}]^{-1}, \quad (22)$$

so a correction splits into an exchange part $Z(\Sigma_x - V_{xc})$ and a correlation part $Z\Sigma_c(E_o)$. With `ExtendOut` every term is printed (columns `Vxc`, `Vnlxc` = Σ_x , `Sc|Eo`, `Re(Z)`), and the equation reproduces the E-Eo column exactly. Columns must be read by their header names: the layout is K-point, Band, Eo, E, E-Eo, ..., and taking the fourth column for $E - E_o$ once produced a false diagnosis (see the errata). `qp_decompose.py` and `cb_splitting.py` read by name.

19.2 WSe₂ without RIM-W

Band edges (`cb_splitting.py`): the gap is *direct* at $\mathbf{K}$, 2.422 eV (VBM band 62 at -1.1195 eV, CBM band 64 at 1.3021 eV); the Q valley (the Γ -K midpoint) lies 0.451 eV above $\mathbf{K}$ (DFT: 0.064 eV), so the indirect $\mathbf{K} \rightarrow \mathbf{Q}$ gap is 2.872 eV. The decomposition, in eV:

difference	total	exchange $Z(\Sigma_x - V_{xc})$	correlation $Z\Sigma_c$
gap at $\mathbf{K}$ (62 $\rightarrow$ 63)	+1.181	+3.135	-1.954
Q against $\mathbf{K}$ (band 63)	+0.368	+0.178	+0.191
band 64 - band 63 at $\mathbf{K}$	-0.055	-0.196	+0.140

The gap behaves normally: correlation closes about 60% of the exchange opening. Two features disagree with the established picture of WSe₂. Both parts push Q up against $\mathbf{K}$, whereas the reference picture has Q at or below $\mathbf{K}$ (an indirect $\mathbf{K} \rightarrow \mathbf{Q}$ gap, the lowest exciton the K-Q state at $\mathbf{M}$). And the conduction spin splitting at $\mathbf{K}$ reverses (+37.2 meV

in DFT, -18.2 meV in GW): exchange stabilises band 64, which has the spin of the valence maximum, whereas the dark exciton of WSe_2 lying below the bright one requires the opposite-spin band lower. With spin-orbit coupling, 200 bands in the response and the self-energy amount to about 100 spinless bands; band convergence, and the lattice constant (3.322 Å against 3.28 Å experimental), are the open suspects [(tests pending)].

19.3 Using corrections in the BSE

A rigid scissor shifts every conduction band equally and keeps DFT's valley offsets and spin splittings: adequate within a valley, wrong between valleys. `KfnQPdb` applies the k -dependent corrections correctly; the check is the report's `[K+QP]` gap, which must equal the Kohn-Sham gap plus the correction at that k . For WSe_2 , RIM-W had to be off in both the GW and the BSE (it removed the binding in the BSE kernel and reduced the GW gap correction at **K** from 1.18 to 0.10 eV); GaN and hBN ran correctly with it.

20. WSe_2 : deferred, and how to resume

WSe_2 is out of the GaN/hBN paper and kept in the package for a later paper on transition-metal dichalcogenides. Paths: `D:\Data\WSe2\` with `LELPH` (`SAVE`, `ph.save`, `ndb.elph`), `GW`, `BSE`, `EXCPH` (`Dmats`, `archives`, `parity` files) and the phonon folder with `copy.sh` and the `wse2_matdyn` inputs.

20.1 What is established

- Spin-orbit symmetry machinery: $\langle D(\sigma_h)^2 \rangle = -1.0000$; band labels from $\pm i$, undetermined only at the Kramers points; phonon parity by permutation ($\pi = [0, 2, 1]$), exact to 7×10^{-16} ; every **Q** = 0 exciton ± 1 .
- Archive symmetry: 1.2×10^{-3} (time reversal), 1.0×10^{-3} (C_3).
- Selection rule: 1.8×10^{-7} and 1.4×10^{-7} at tolerance 0.96 ; 6.3×10^{-6} with manifold rotation (state 1). The residue at the default tolerance is one solver-mixed pair; 25% of manifold eigen-parities stay short of ± 1 for a reason not yet found (state cut and manifold width ruled out; window closure to test).
- For WSe_2 , RIM-W must be off in the GW and the BSE (it works for GaN and hBN; the cause is not isolated); `KfnQPdb` works.

20.2 What is open

- The GW (no RIM-W, 200 bands, $a = 3.322$ Å, 12×12): direct gap at **K** (2.42 eV), **Q** 0.43 – 0.45 eV above **K**, conduction spin splitting at **K** reversed ($+37$ meV in DFT, -18 meV after GW).
- Two published pictures. Mishra *et al.* [PRB **98**, 045143 (2018)]: fully relativistic ONCV with `W 5s5p4f` semicore, PBE, $a = 3.3175$ Å, 18×18 DFT mesh; G_0W_0 gap 2.19 eV, direct; **Q** lifted from 44 to 394 meV above **K** (as here); conduction splitting at **K** kept at $+40$ meV. Chen, Sangalli and Bernardi [PRR **4**, 043203 (2022)]: $a = 3.27$ Å; exciton minima at **M** and **Q**.
- Reading: the **Q**–**K** offset agrees with Mishra *et al.*; the difference from Bernardi is mainly the lattice constant (compressive strain lowers **Q** relative to **K**, about 100 meV per % in the bilayer), i.e. pseudopotential and functional. The reversed conduction splitting is ours alone: an exchange effect ($Z\Delta\Sigma_x = -0.196$ eV between the spin partners) to be checked against the `W` semicore and the Σ_x cutoff. The gap (2.42 against 2.19 eV) is a mesh question.
- Consequently the exciton landscape, the archives and everything downstream are not yet trustworthy for WSe_2 .

20.3 Resume, in order

- Pseudopotential and exchange** (cheap, settles the conduction order): check the `W` valence configuration in the UPF header (`5s5p` and `4f` semicore?); run Hartree-Fock only (`yambo -x`) at $k = 19$, bands 62 – 64 , with `EXXRLvcs` at 70 , 100 and 140 Ry and watch $\Sigma_x(64) - \Sigma_x(63)$.
- Lattice constant**: DFT bands at $a = 3.27$ and 3.28 Å (`scf + nscf` only) to see how far **Q** moves; G_0W_0 at **Q** and **K** only if it changes the picture.
- GW band convergence** at $k = 13$ (**Q**) and 19 (**K**) only: `QPkrange 13|13|62|64|` and `19|19|62|64|`; `BndsRnXp` and `GbndRnge` at 200 , 400 , 600 (spinor bands; `nscf` with enough bands); `rim_cut` on, `rim_w` off. After each: `cb_splitting.py` (**Q**–**K** offset; 64 – 63 at **K**) and `qp_decompose.py`.
- Mesh**: G_0W_0 on 18×18 (**Q** and **K** are mesh points) for the gap.
- Full GW** at the converged settings, all 19 k , bands 59 – 66 .
- BSE** with `KfnQPdb = "E < GW/ndb.QP"`, no `rim_w`, `BSSmod = "d"`, all 19 **Q**. Check the report's `[K+QP]` gap against `cb_splitting.py`; `exc_transitions.py` at **Q** = **M** (`ndb.BS_diago_Q7`): is the lowest state a **K**– Λ exciton?
- Symmetry**: `xphd parity -manifold-tol 0.005; window_closure.py (SIGMA_H_OP = 6); manifold_closure.py`.
- Archives and analysis**: `xphd dmats`; `xphd generate` (states enough to close the manifolds); `check-archive`; `labels_from_elph.py`; `selection-rule -rotate`; then `linewidths`, `add_osc.py`, the BTE and the maps as in Part I, Sec. 9.2 (read with the errata).

21. Validation summary

Check	Result	Where
Exciton parity, $ \chi = 1$	100% of states (GaN)	Sec. 14.4
Selection rule, forbidden/allowed	5×10^{-7} (GaN); 1.8×10^{-7} (WSe ₂ , tol 0.96)	Sec. 14
Star uniformity $T_{\max}/T_{\min}$	1.0001 (GaN)	Sec. 13
Manifold-resolved symmetry, TR / C ₃	$1.3/3.3 \times 10^{-3}$ (GaN); $1.2/1.0 \times 10^{-3}$ (WSe ₂)	Sec. 13
Phonons, matdyn vs source mesh	1.7 μ eV	Sec. 16
K minimum vs closed form, 77 K	10.4 vs 10.6–11.4 meV	Sec. 16
Off-diagonal self-energy	< [0.3] % of the lineshape	Sec. 16
Helicity sum rule; P per valley	3.7×10^{-16} ; ∓ 1.0000	Sec. 17
BTE row sums vs linewidths	< 0.1%	Sec. 18
BTE detailed balance; stationarity	machine precision; 10^{-16}	Sec. 18
BTE end state vs Boltzmann, 77 K	4.26 vs 4.09 meV	Sec. 18
Spinor mirror, $\langle D(\sigma_h)^2 \rangle$	−1.0000 (WSe ₂)	Sec. 14
Phonon parity by permutation	± 1 to 7×10^{-16} (WSe ₂)	Sec. 14

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